

Fifty Shades of Green: Central Bank Communication about Climate Change and Inflation Expectations

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ABSTRACT

Central banks increasingly communicate about climate change, yet the implications for market-based inflation expectations remain unclear. We compile a new dataset of climate-related speeches by European Central Bank and euro-area national central bank officials from 2008–2025. Using local projection regressions on daily euro-area inflation-linked swap rates (1-, 2-, 5-, and 10-year maturities), and controlling for euro-area macroeconomic surprises, ECB meeting days, and non-climate speeches delivered on the same day, we estimate the dynamic response of inflation expectations to the frequency of climate speeches and the intensity of climate-related language. We find that climate-speech frequency and intensity are associated with a modest and delayed increase in inflation-linked swap rates. We also document limited speaker heterogeneity: market reactions are, on average, stronger for Executive Board members and officials from founding euro-area countries, but are confined to particular maturities. A channel decomposition indicates that mandate-drift tone and inflation-risk framing raise inflation expectations, whereas mandate-affirming, price-stability-oriented (hawkish) climate communication offsets upward pressure. Overall, the estimated effects are statistically detectable but economically small, consistent with climate communication affecting pricing at the margin without evidence of de-anchoring of inflation expectations.

Highlights

- We compile a new dataset of European Central Bank and euro-area national central bank climate speeches (2008–2025) and link it to market-based inflation expectations.
- Climate communication is followed by a modest and delayed rise in inflation-linked swaps.
- Channel evidence points to offsetting effects: mandate-drift and inflation-risk framing push inflation expectations up, while mandate-affirming price-stability language tempers the increase.
- Climate communication increases forward volatility of long-term inflation swaps, consistent with heightened uncertainty about the monetary policy reaction function.
- Speaker heterogeneity is limited; effects are slightly larger for Executive Board and founding-member officials, and are maturity-specific.
- Overall effects are statistically detectable yet economically small, consistent with anchored inflation expectations in the euro area.

Keywords: Central bank communication; Climate change; Inflation expectations; Central bank mandate; Monetary policy; Local projections; European Central Bank.

JEL Classification

E52; E58; E31; Q54; C32.

1. Introduction

An increasing number of political leaders are calling for central banks to take a more active stance in mitigating climate-related risks. For example, at Jerome Powell's Nomination Hearing (2022), US Senator Elisabeth Warren remarked: "The world is running out of time to deal with the climate crisis, and the Fed has an important role to play here, and I hope the Fed will step up." Many central banks have accepted responsibility for fighting climate change. By introducing the first comprehensive climate stress test for banks and insurance firms in 2020, the Bank of France took a pioneering step in addressing climate-related financial risks, setting a precedent for central bank action in this area. In its 2021 monetary policy strategy review, the Governing Council of the European Central Bank (ECB) acknowledged the importance of climate change for monetary policy, leading to the adoption of a concrete climate action plan. As ECB President Christine Lagarde said at the 2021 International Climate Change Conference: "In short, climate change has consequences for us as a central bank pursuing our primary mandate of price stability, and our other areas of competence, including financial stability and banking supervision."¹

Recognizing the significance of climate-related risks and opportunities, central banks and supervisors established the Network for Greening the Financial System (NGFS) in 2017 with the purpose of "strengthening the global response required to meet the goals of the Paris Agreement and enhancing the role of the financial system in managing risks and mobilizing capital for green and low-carbon investments."² However, central banks face a challenge of how to integrate climate considerations into their monetary policy frameworks without compromising their

¹ "Climate Change and Central Banks: Analysing, Advising and Acting," Speech by Christine Lagarde, President of the ECB, at the International Climate Change Conference in Venice, Venice, 11 July 2021.

² See: <https://www.ngfs.net/en/about-us/origin-and-purpose#:~:text=To%20carry%20its%20ambition%20as,analytical%20work%20on%20green%20finance>.

primary objective, price stability.³ Benoît Cœuré, a then member of the ECB's Executive Board, argued that climate change could "complicate the correct identification of shocks relevant for the medium-term inflation outlook, it may increase the likelihood of extreme events and hence erode central banks' conventional policy space more often, and it may raise the number of occasions on which central banks face a trade-off forcing them to prioritize stable prices over output."⁴

How can central banks maintain monetary credibility while addressing climate change, which extends beyond their traditional mandate of price stability? Central bank communication has become a core instrument for strengthening the predictability and credibility of monetary policy (Arseneau, Drexler, and Osada, 2022). Its importance has increased substantially over the past decade as policy goals and implementation tools, including unconventional monetary policy tools, have become more complex. As Blinder et al. (2008: 910) argue, central bank communication "has the ability to move financial markets, to enhance the predictability of monetary policy decisions, and potentially to help achieve central banks' macroeconomic objectives." Through speeches, press conferences, and other public statements, central banks shape market perceptions not only of their intentions but also of their broader policy priorities.

And yet, the effects of climate-related communication, particularly its influence on inflation expectations, are still not well understood. This study is, to the best of our knowledge, the first to examine how central banks' communication on climate change influences the inflation expectations of financial market participants. To achieve this objective, we compiled a novel dataset of speeches delivered by the representatives of the ECB and national central banks

³ Many central banks are expected to align with their governments' climate and environmental goals, although few have explicit sustainable development mandates (Dikau and Volz, 2021).

⁴ "Monetary policy and climate change," Speech by Mr Benoît Cœuré, Member of the Executive Board of the ECB, at a conference on "Scaling up Green Finance: The Role of Central Banks", organized by the Network for Greening the Financial System, the Deutsche Bundesbank and the Council on Economic Policies, Berlin, 8 November 2018.

in eurozone countries on climate change from January 1, 2008, to December 31, 2025, coding both the frequency and intensity of climate-related language. We hypothesize that climate-related communication affects inflation expectations primarily by increasing uncertainty about the central bank's mandate, i.e., concerns about "mission creep," and about future policy changes. Employing the local projections method (Jordà, 2005), we examine the dynamic effects of climate speeches on market inflation expectations across different maturities, as measured by inflation-linked swaps. Our analysis of daily inflation-linked swaps (ILS) across 1, 2, 5, and 10-year maturities indicates that the frequency and intensity of central bank speeches are associated with a positive, though modest, upward shift in inflation expectations, and that this impact tends to manifest with some delay. These effects persist even when we account for euro area macroeconomic surprises, ECB Governing Council meetings and decisions, alternative estimation techniques, and an extensive set of control variables. The study yields the following main findings: First, central bankers' climate-related speeches exert statistically detectable yet modest effects on financial market participants' inflation expectations, while not generating evidence of de-anchoring. Second, the effects appear to run through two conflicting mechanisms. Communication highlighting mandate drift or inflationary consequences of climate disruptions tends to raise inflation compensation, whereas mandate-affirming communication that reinforces the primacy of price stability offsets this upward pressure and helps keep expectations anchored. This suggests that the ECB can *neutralize* inflation expectations associated with climate policies by reinforcing its commitment to the mandate of price stability. Third, impulse-response evidence indicates modest speaker heterogeneity: speeches delivered by central bankers from founding eurozone countries and ECB Executive Board members show somewhat larger

responses than those by the ECB President, though these differences are concentrated at specific maturities.

This paper makes contributions to three distinct areas of scholarship. First, it contributes to growing research on central bank communication.⁵ Central banks have transitioned from a secretive approach to a transparent communication with the markets and the public. A notable example is the former Fed governor Alan Greenspan, renowned for his enigmatic communication style, humorously labeled as "Fed speak." This "communication revolution" has led central banks to conduct press conferences and deliver speeches after policy decisions, explaining their present strategies and future trajectories. Scholars have found that central bank communication shapes households' consumption and saving behavior (Coibion, Gorodnichenko, and Weber, 2022), as well as financial markets and asset prices (Gürkaynak, Sack, and Swanson, 2005; Blinder et al., 2008; Swanson, 2021; Gertler and Horvath, 2018; Ehrmann and Fratzscher, 2007; Nakamura and Steinsson, 2018; Hansen and McHanan, 2016; Altavilla et al., 2019; Ambler and Rumler, 2019; Picault, Pinter, and Renault, 2022). Other studies explored *non-monetary* communication, recognizing that central banks address a broad range of topics beyond traditional monetary policy issues (e.g., economic growth) (e.g., Cieslak and Schrimpf, 2019; Moschella and Pinto, 2019; Hansson, 2021). Despite the expanding body of research on central bank communication, climate-related communication remains relatively underexplored. Notable exceptions include recent studies by Arseneau, Drexler, and Osada (2022), Arseneau and Osada (2023), and Campiglio et al. (2025) that examine *how* central banks address climate change and how climate narratives have evolved.

⁵ For excellent reviews of the literature, see, for example, Blinder et al. (2008) and Blinder et al. (2024).

Second, our paper connects with the broader literature on the central bank mandate (e.g., Bohl, Kanelis, and Siklos, 2023). There is no consensus in the literature and policy circles regarding the desirability of central banks' involvement in addressing climate change. Some scholars argue that climate change considerations should be integrated into monetary policy frameworks (McKibbin et al., 2021). Others caution, however, that pursuing climate objectives directly could erode central bank credibility, and argue that climate goals are better advanced through fiscal, rather than monetary, policy (Hansen, 2022).⁶ Inadequate government action to ensure a successful low-carbon transition could, in turn, constrain central banks' ability to fulfill their price-stability mandate (Bolton et al., 2020). The analysis of the ECB's climate-related speeches is particularly relevant because, even though it does not fall explicitly within the ECB's primary mandate, climate change is increasingly recognized as a systemic risk. As highlighted by Fabio Panetta, a Member of the ECB's Executive Board, during a session with the Italian Banking Association in 2022: "Given that the ECB's primary mandate is to preserve price stability, understanding the relationship between the transition to a greener economy and the price of energy is crucial."⁷

Finally, this study relates to the scholarship on inflation expectations.⁸ Inflation expectations play a central role in the transmission of monetary policy, as they influence how households and firms determine wages and prices, which in turn impact actual inflation (Bernanke et al., 2001). As monetary policy has shifted towards a more forward-looking

⁶ Governments can employ fiscal policies to mitigate climate change and support a shift towards a low-carbon economy. This includes implementing carbon pricing mechanisms, such as taxes or emissions trading systems, but also public investments in climate-resilient infrastructure (e.g., early warning systems and disaster preparedness). See, for example, Pigato (2019).

⁷ Fabio Panetta, "Greener and cheaper: Could the transition away from fossil fuels generate a divine coincidence?" Speech at the Italian Banking Association, Rome, 16 November 2022.

⁸ For recent surveys of the determinants and evolution of inflation expectations, see Cieslak and Pflueger (2023), Kose et al. (2019).

approach, expectations regarding future policies have gained significance. Researchers have examined how households, firms, and professional forecasters form their expectations about future price dynamics (Coibion et al., 2018; Coibion et al., 2020; D'Acunto et al., 2021; Binder and Kamdar, 2022), while Gürkaynak et al. (2010) and Beechey et al. (2011) explore market-based inflation expectations.⁹

This paper makes two novel contributions. First, we compile an expanded dataset of the speeches by ECB and eurozone central bank officials on climate change through 2025 that distinguishes speakers by role (e.g., presidents, executive board members), enabling a more granular analysis of both the evolution and heterogeneity of climate-related communication. Second, to our knowledge, we are the first to examine whether and how this climate communication affects financial markets' perceptions of inflation.

The rest of the paper is organized as follows. Section 2 describes our hypotheses on how central bank communication could impact inflation expectations. Section 3 introduces our methodology, and provides a descriptive analysis of our sample of climate-related speeches. Section 4 presents the main results and extensions, Section 5 carries out a set of robustness checks, and Section 6 concludes.

2. Central Bank Communication Channels and Inflation Expectations

When a central bank addresses climate change, it highlights the significance of climate-related factors in the economy. By doing so, the bank can construct "narratives" that prepare the markets

⁹ Coibion et al. (2020) and others reveal that in low inflation environments, central bank communication doesn't specifically target households and firms and their inflation expectations don't show significant responses to monetary policy announcements.

for potential monetary policy and regulatory changes.¹⁰ Given the informational asymmetry between the public/markets and the central bank, communication plays a crucial role in alleviating information problems and promoting investor learning. This is important because, as Fisher (1911) argued (cited in Binder and Kamdar, 2022), cognitive limitations and inattention may hinder the average businessperson, who often lacks strong theoretical knowledge, from forming accurate inflation expectations. By improving knowledge, central bank communication can help reduce the gap between expected and realized inflation. Such communication serves two main purposes: first, it conveys signals about the private information held by central banks; and second, it acts as a coordination tool that aligns the beliefs of financial market participants (Amato, Morris, and Shin, 2002)

Our main prediction is that climate-related central bank communication influences market-based inflation expectations by increasing *uncertainty* over whether the bank will adhere strictly to price stability or compromise it to advance climate goals. This ambiguity, known as "mission creep," is linked to potential mandate expansion and tradeoffs between climate objectives and inflation control. When there is a consensus regarding the central bank's long-term inflation objective, inflation expectations remain firmly anchored; when there is no such consensus, expectations become more susceptible to revision in response to incoming news (Beechey, Johannsen, and Levin, 2011). We use the frequency and intensity of climate-related language in central bank speeches as proxies for the salience of climate considerations in communication. Greater salience may affect inflation expectations by increasing uncertainty about future policies when climate communication is interpreted as signaling new tradeoffs with

¹⁰ For the role of economic narratives, see Shiller (2017) and narratives related to central bank communication (Hansson, 2021).

the price-stability mandate, but also potentially reducing uncertainty when it clarifies that climate issues will be addressed consistently with that mandate.

Therefore, we assume that the effect is heterogeneous by the content of communication. In our framework, climate-related central bank speeches are a source of uncertainty that can affect market-based inflation expectations through three distinct channels. The first source of uncertainty is that central banks' involvement in climate policy may lead to "mandate-drift," when climate objectives undermine the *mandate* of price stability. When central banks communicate about climate risks, financial markets may grow uncertain about how these monetary authorities will prioritize between their core objective of maintaining price stability and supporting the green transition. Markets may expect central banks to adopt expansionary monetary policies to support climate goals or mitigate the inflationary impact of disasters, employing interest rate adjustments, targeted credit programs (e.g., green lending), or green quantitative easing. For instance, as part of its climate roadmap to promote green financial investments and lower funding costs for clean-energy companies, the ECB incorporated climate considerations into its existing asset-purchase framework by tilting corporate-bond reinvestments toward issuers with better climate performance in October 2022.¹¹ Strong demand for environmental projects, however, could lead to investment-driven inflation, especially within green economic sectors.

Incorporating climate objectives into monetary policy also introduces uncertainty in the monetary-fiscal policy mix. Central banks might consider combating climate change using prudential ratios in the banking sector to influence credit distribution, possibly leading to a shift towards "directed credit" policies (Brunnermeier and Landau, 2020). However, if central banks

¹¹ <https://www.ecb.europa.eu/press/pr/date/2022/html/ecb.pr220704~4f48a72462.en.html>

intervene to accelerate the green transition by favoring certain sectors through favorable lending rates or specific loan programs, or investing their reserves in green assets without adherence to traditional principles of liquidity and safety, they risk being perceived as overstepping their legal mandate (Tamez, Weening, and Yoshinaga, 2024). Moreover, transferring climate-related fiscal policy responsibilities to central banks risks subjecting monetary policy to political pressures and lobbying pressures (Hansen, 2022). Different sectors, regions, and firms would try to influence climate-related financial regulations or asset purchase programs to their advantage. Such political interference could undermine the independence of central banks and weaken their capacity to maintain price stability, which may lead financial markets to adjust their inflation expectations upward.

Climate communication may thus complicate the ability of market participants to form stable and accurate inflation expectations, given that climate change lies beyond the traditional mandate and expertise of central banks. If financial markets interpret climate-related speeches as signaling that the central bank is willing to tolerate higher inflation and will be hesitant to raise interest rates to support the green transition, then perceived mandate drift can push inflation expectations upward.

Mandate Drift Hypothesis: Mandate-related tone in climate communication is likely to be associated with higher inflation expectations.

A second channel works through new information in central bank communication about how climate change and the green transition could drive inflation. Climate shocks and decarbonization policies can contribute to higher inflation over a longer horizon. During the transition to a low-carbon economy, companies may face higher energy input costs from carbon pricing that raises fossil fuel prices, alongside upfront investments required for cleaner

technologies. Higher interest rates may slow the green transition by increasing the cost of capital for renewable technologies, while indirectly protecting incumbent oil and gas producers from new low-carbon competitors.¹²

Empirical work also suggests that natural disasters and weather shocks (such as droughts) can negatively impact agricultural production, leading to supply shocks and higher food prices (Dikau and Volz, 2021). For instance, Kotz et al. (2023) found that global food price inflation could rise by 1 to 3 percentage points annually by 2035, contributing an estimated 0.3 to 1.2 percentage points to headline inflation. Beirne et al. (2024) show that weather-related disasters increase euro-area inflation mainly by intensifying adverse supply shocks (e.g., crop and infrastructure damage) and generating positive demand shocks (e.g., reconstruction). They also show that these disasters increase prices disproportionately for goods that make up a larger share of low-income households' spending, thus worsening inflation inequality. Higher energy costs resulting from weather-induced disruptions to commodity supply (e.g., floods or storms) can lead to cost-push inflation, in which businesses pass higher production costs on to consumers in the form of higher prices (Batten, Sowerbutts, and Tanaka, 2020).¹³ Natural disasters can also reduce economic output by disrupting activity, which in turn lowers household consumption and firms' investment and weakens overall business confidence (Batten, Sowerbutts, and Tanaka, 2016).

Climate-related speeches by central banks can provide information about how physical risks (e.g., climate disasters such as floods, storms, or wildfires) and transition risks (e.g., carbon pricing, stricter regulation, and stranded assets) may affect inflation dynamics. Isabel Schnabel, a member of the ECB's Executive Board, drew attention to inflationary pressures linked to

¹² Thomas Ferguson and Servaas Storm. "Central banks raising interest rates makes it harder to fight the climate crisis." *The Guardian*, May 6, 2023.

¹³ We thank Oleg Itshoki for pointing to the importance of cost-push shocks.

climate-related events and coined the term 'climateflation' to capture their potential to generate persistent price pressures.¹⁴ Reflecting this concern, Banque de France Governor François Villeroy de Galhau has described global climate shocks linked to surging energy prices as a “stagflationary shock” that would mean persistently higher inflation combined with slower economic growth.”¹⁵

Climate-Inflation Information Hypothesis: More inflation-risk content in climate communication is likely to be associated with higher inflation expectations.

A third channel highlights credibility rather than mandate drift: climate-related speeches may *reduce* uncertainty by reassuring markets that climate issues will be handled within the central bank’s existing mandate (i.e., price stability remains the primary objective). Investors likely pay attention not only what topics the central bank discusses, but also what that communication means for the priority given to price stability and the policy trade-offs the central bank is willing to make. When a climate-related speech reiterates commitment to the inflation target, it can lower perceived mandate drift risk and thus anchor inflation expectations. This motivation is consistent with evidence that central bank mandates influence the content and tone of communication (Bohl, Kanelis, and Siklos, 2023) and that ECB communication can be quantified along stance dimensions (hawkish vs. dovish) (Tobback, Nardelli, and Martens, 2017). Even climate-salient speeches need not unanchor inflation expectations if they reaffirm the commitment to price stability. For example, Isabel Schnabel, a member of the ECB’s Executive Board, explicitly framed the ECB’s climate-related considerations as fully consistent

¹⁴ Isabel Schnabel, “A new age of energy inflation: Climateflation, fossilflation and greenflation,” speech at the panel “Monetary Policy and Climate Change”, The ECB and its Watchers XXII Conference, Frankfurt am Main, 17 March 2022.

¹⁵ Francois Villeroy de Galhau, Governor, Banque de France, Keynote Address at the World Conference of Banking Institutes: The Role of Banking in a Sustainable Global Economy 6 (September 17, 2019).

with its primary objective, arguing that “failing to arrest high inflation in a timely manner would jeopardise the green transition,” and that a restrictive stance is needed to “restore price stability” and keep longer-term inflation expectations anchored.¹⁶

Mandate-Affirming Hypothesis: Price stability reaffirming climate communication should be associated with lower (or stable) inflation expectations.

3. Data and Methodology

3.1. Central Bank Communication Dataset

This paper uses a collection of central bank speeches from the Bank for International Settlements, spanning from January 1973 to the present, which includes over 18,000 speeches delivered by representatives of more than 100 central banks. We focus on central bankers’ speeches rather than minutes or official statements because speeches deliver real-time, publicly accessible information and occur far more frequently than written disclosures such as meeting minutes and reports. Speeches are also more informative than formal announcements, since they often cover a wider range of topics, and their importance within central banks’ overall communication strategies has expanded significantly (Hansson, 2021). Speeches serve as an important vehicle for communicating monetary policy objectives (Gürkaynak, Sack, and Swanson, 2005), since they reflect direct remarks from individual members of the policy-making committee and therefore let us differentiate across speakers, follow how their positions change over time, and assess how these messages shape financial markets and public perceptions.

Our primary focus is on those addressing climate change. To identify speeches as

¹⁶ “Monetary policy tightening and the green transition,” Speech by Isabel Schnabel, Member of the Executive Board of the ECB, at the International Symposium on Central Bank Independence, Sveriges Riksbank, Stockholm, 10 January 2023.

"climate-related," we adopted a methodology similar to that used by Arseneau, Drexler, and Osada (2022). Searching for the keyword "climate change" initially produced 2,223 speeches without country filters. Subsequently, we refined our dataset to include only speeches from ECB and eurozone national central bank officials, delivered between January 1, 2008 and December 31, 2025.

We further refined this initial set of speeches through an iterative process using climate change-related keywords identified by Arseneau, Drexler, and Osada (2022). The keywords used to refine our dataset included terms such as "Green Finance," "Climate-related Risk," "Paris Agreement," "Climate Policy," "Climate Risk," "Low-carbon Economy," "Carbon Emission," and "Green Bond."¹⁷ Our refinement process was guided by a rule stipulating that a speech must contain any keyword in at least two separate sentences within the speech to be included in the final set of climate-related speeches. This process resulted in 514 speeches delivered by representatives of the ECB and members of the national central banks from the eurozone (across 402 days).¹⁸

In addition to the *count* indicator that measures the daily frequency (number) of climate-related speeches, we also construct a climate speech *score*, which represents the share of climate-related words to the total number of words, indicating the intensity of speeches. These two

¹⁷ To create a consistent measure of the ECB, we made several decisions. First, we recorded the dates when the speeches were delivered, not when they were reported in the BIS database. Second, if a meeting or a conference lasted two days or an interview was broadcast, we recorded it in our database on the first day. It's worth noting that our database contains only one meeting that spanned two days: the speech by Ignazio Visco, Governor of the Bank of Italy at the Rome Roundtable 2020, a virtual meeting organized by The Global Foundation on "The G20 under Italy's leadership in 2021", taking place on November 16-17, 2020. After reviewing the program of the Roundtable, we determined that the speech was delivered on November 17, 2020, and therefore used this date. Third, for interviews published in newspapers, we coded the publication date rather than the date the interview was conducted.

¹⁸ The coding of speeches was carried out independently by research assistants and one of the authors, and subsequently validated and unified by another author.

variables capture the frequency and intensity of climate-related discourse over time. Like Ambler and Rumler (2019), we also include a series of dummy variables related to the speaker's identity. Specifically, we create dummy variables to identify speeches delivered by the President of the ECB, members of the Executive Board of the ECB,¹⁹ and central bankers from founding member countries of the eurozone²⁰. This allows us to test hypotheses regarding whether financial markets react differently to communication from these speakers.

4.2. Description of Central Bank Speeches



Figure 1: Total number of climate-related speeches by the ECB and national central banks, 2008-2025.

As depicted in Figure 1, based on our dataset, the climate-related speeches are delivered mainly by the ECB representatives – the President of the ECB and members of the ECB Executive Board (122 speeches) – followed by central bankers from Germany (79 speeches),

¹⁹ Note that the Executive Board of the ECB consists of the President, the Vice-President, and four other members.

²⁰ There are 11 founding members of the eurozone: Austria, Belgium, Finland, France, Germany, Ireland, Italy, Luxembourg, the Netherlands, Portugal, and Spain.

France (57), Ireland (47), Spain (44), Italy (43), Netherlands (38), Finland and Greece (both 20 speeches). In contrast, representatives from smaller central banks, such as those of Luxembourg, Estonia, Belgium, and Malta, discuss climate change relatively infrequently in absolute terms.

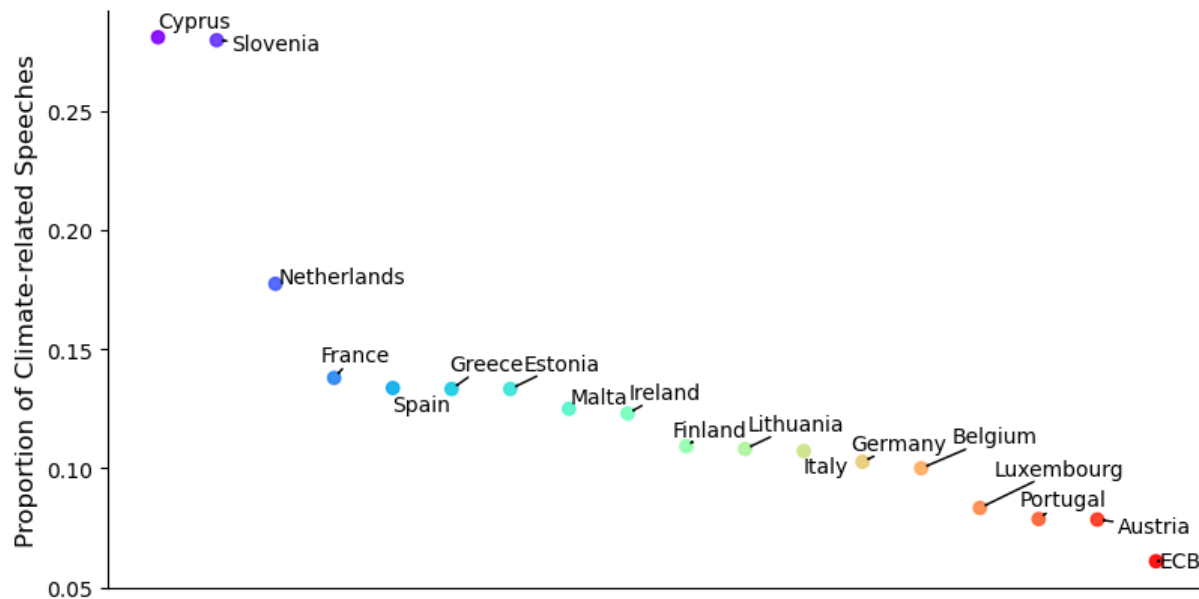


Figure 2: Climate-related speeches as a percentage of total speeches by the ECB and national central banks, 2008–2025.

Even so, as Figure 2 shows, only 6.1% of all speeches given by ECB representatives were about climate change. Low shares are also observed among representatives of the central banks of Austria (7.8%), Portugal (7.9%), and Luxembourg (8.3%). In contrast, central bank representatives from the Netherlands discuss climate change relatively frequently as a proportion of their total speeches, along with their counterparts from Slovenia and Cyprus.

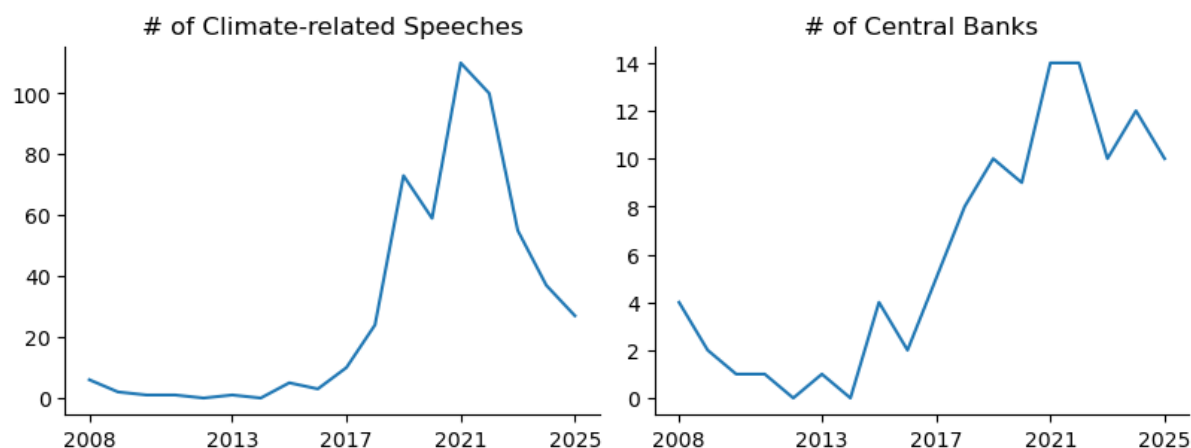


Figure 3: Climate-related speeches over time and number of central banks communicating about climate change, 2008-2025.

Figure 3 illustrates a growing trend in central bank speeches mentioning climate change in recent years, notably accelerating after Mark Carney's influential 2015 speech, "Breaking the Tragedy of the Horizon: Climate Change and Financial Stability."²¹ Between 2018 and 2022, communication on climate change became particularly prominent, with a significant rise in the number of central banks and their representatives addressing climate change issues. However, our data show that communication intensity peaked in 2022 and then stabilized rather than continued to rise sharply.²²

²¹ "Breaking the tragedy of the horizon – climate change and financial stability," Speech by Mr Mark Carney, Governor of the Bank of England and Chairman of the Financial Stability Board, at Lloyd's of London, London, 29 September 2015.

²² This pattern aligns with institutional developments inside the ECB. The establishment of the Climate Change Centre in 2021 represented a dedicated governance structure for climate policy, and was later consolidated in the 2024–2025 "climate and nature plan," embedding climate considerations across monetary policy, banking supervision, data, and market infrastructure (ECB Economic Bulletin, Issue 2/2024). The ECB's communication also changed from signaling and agenda-setting toward implementation of supervisory stress tests and disclosure requirements (see ECB 2022). As a result, speeches from 2023 onward tended to emphasize progress in banks' climate-risk management rather than new announcements. In parallel, the ECB tried to lower its operational carbon footprint by scaling back physical meetings, events, and conferences, thereby mechanically limiting opportunities for separate climate speeches. As a result, climate-related communication stabilized and consolidated rather than continuing its rapid expansion. Finally, the 2022 war in Ukraine plausibly redirected attention toward energy shocks and inflation, contributing to a flattening of "green intensity" in ECB communication (see European Central Bank, 2024 update of the ECB's Environmental Statement).

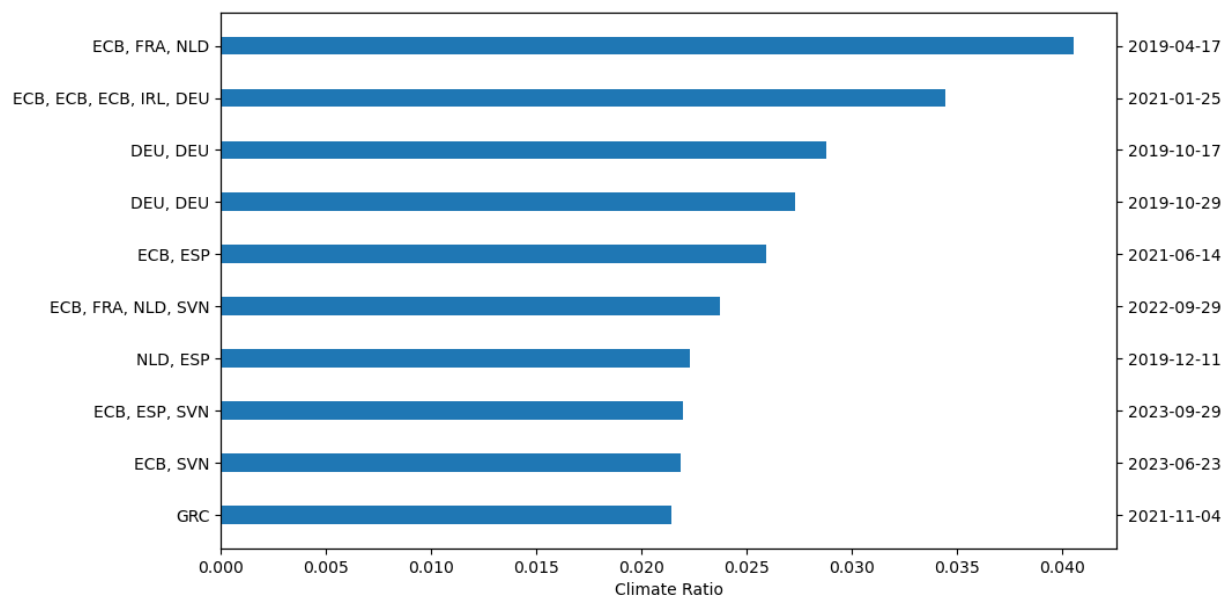


Figure 4: Most relevant climate-related speeches: 10 dates with the highest climate speech scores, 2008-2025.

We further identify the most significant speeches to trace the evolution of climate-related discourse across central banks over time. Figure 4 depicts the ten dates with the highest "climate speech scores" (climate intensity) measured as the share of climate-related words relative to the total word count. The figure indicates that on April 17, 2019, central bank officials from the ECB, France, and the Netherlands delivered speeches with the highest climate intensity. The second most significant cluster of speeches occurred on January 25, 2021, when ECB representatives, along with central bankers from Ireland and Germany, emphasized climate issues in their speeches. Overall, central bankers gave the most climate-focused speeches between April 2019 and September 2023. This trend is consistent with the prominent role of eurozone central banks in advancing climate-related initiatives since the creation of the NGFS. The French, Dutch, and German central banks were among the group's eight founding members. The Banque de France is playing a particularly pivotal role, having first proposed the NGFS in 2017 and subsequently hosting its permanent secretariat. This trend is consistent with the

prominent role of eurozone central banks in advancing climate-related initiatives since the creation of the NGFS.

Table 1 illustrates that the most significant climate-related speeches were delivered at the NGFS Conference on April 17, 2019, by François Villeroy de Galhau, Governor of Banque de France; Frank Elderson, Executive Director at the Dutch Central Bank; and Sabine Lautenschläger, a member of the ECB's Executive Board. The ECB's increased focus on green policy during 2019–20 was driven by personnel and leadership changes within the institution (Deyris 2023). During this time, new appointments to the Executive Board, such as Frank Elderson and Isabel Schnabel, became key figures advocating for green policies that went beyond just addressing financial risk. Elderson played a crucial role in incorporating climate change considerations into central banking as the first chair of the NGFS from its establishment in 2018 until 2022. Under his leadership, the NGFS evolved into a key global coalition of central bankers and financial regulators addressing financial risks related to climate change. Elderson's influence significantly grew in 2020 when he became the co-chair of the Task Force on Climate-related Financial Risks, operating under the Basel Committee on Banking Supervision. This position allowed him to play a crucial role in integrating climate risk management into the global banking sector.

Deyris (2023: 717) notes that by 2021, references to climate change in ECB speeches had reached the same frequency as mentions of inflation. The ECB's early interest in climate change stemmed from concerns that it could pose systemic financial risks, a concern shared by many central bankers following the 2008 financial crisis.²³ However, the ECB officials have

²³ We are grateful to Michael Ehrmann for this insight.

increasingly examined the implications of climate change for other aspects of their work, such as monetary policy and macroeconomic forecasting. As German Bundesbank President Jens Weidmann stated: "The importance of climate change and climate policy for central banks is not confined to our role as financial supervisors and guardians of financial stability, though. There can be all manner of monetary policy implications, too. Both climate change and climate policy can have a bearing on macroeconomic indicators such as output and inflation."²⁴

²⁴ "Climate change and central banks," Welcome address by Dr Jens Weidmann, President of the Deutsche Bundesbank and Chairman of the Board of Directors of the Bank for International Settlements, at the Deutsche Bundesbank's second financial market conference, Frankfurt am Main, 29 October 2019.

Table 1: Top Climate-Related Speeches

Date	Country	Speaker	Position	Title
2019-04-17	France	François Villeroy de Galhau	Governor	Central banks and supervisors taking action on climate change at the NGFS conference.
	Netherlands	Frank Elderson	Executive Director	Network for Greening the Financial System's recommendations for policymakers, central bankers, and supervisors, at the NGFS conference.
	ECB	Sabine Lautenschläger	MEB	Central bankers, supervisors and climate-related risks, at the NGFS conference.
2019-10-17	Germany	Joachim Wuermeling	MEB	A new world ahead. What do sustainable finance and digitalisation mean for supervision? at the occasion of the ESE Conference.
	Germany	Sabine Mauderer	MEB	Opening remarks at the US Senate Democrats' Special Committee on the climate crisis.
2019-10-29	Germany	Jens Weidmann	President	Climate change and central banks, at the Deutsche Bundesbank's second financial market conference.
	Germany	Sabine Mauderer	MEB	Central banks - a crisis manager for the climate? At the Deutsche Bundesbank's second financial market conference.
2021-01-25	ECB	Christine Lagarde	President	Climate change and central banking, at the ILF conference on Green Banking and Green Central Banking.
	ECB	Fabio Panetta	MEB	Sustainable finance - transforming finance to finance the transformation, at the 50th anniversary of the Associazione Italiana per l'Analisi Finanziaria (virtual).
	ECB	Frank Elderson	MEB	Introductory Statement, Hearing at the Committee on Economic and Monetary Affairs of the European Parliament.
	Ireland	Gabriel Makhoul	Governor	The year ahead at the University of Limerick.
	Germany	Jens Weidmann	President	What role should central banks play in combating climate change? At the ILF conference on Green Banking and Green Central Banking.
2021-06-14	ECB	Isabel Schnabel	MEB	From market neutrality to market efficiency, at the ECB DG-Research Symposium "Climate change, financial markets and green growth."
	Spain	Pablo Hernández de Cos	Governor	What will be the impact of climate transition on monetary policy? At the Central Banking Summer Meetings.

3.2. *Inflation Expectations*

We measure financial market inflation expectations using daily data on inflation-linked swap rates (spot rates) for euro area inflation with maturities of 1, 2, 5, and 10 years (see Gürkaynak et al., 2005; Beechey, Johannsen, and Levin, 2011).²⁵ Inflation-linked swaps are financial derivatives to manage or hedge against rising inflation risks. The buyer of the inflation swap pays a fixed rate corresponding to the anticipated inflation at the contract's maturity date. The inflation swap market is the platform where investors trade "inflation risk." Prices in the inflation swap market are closely watched because they reveal market-based inflation expectations, often summarized in the breakeven inflation rate.

The euro-area inflation swaps market is the primary market for hedging exposures indexed to consumer prices in the euro area. The participants in this market include institutions with inflation-linked liabilities, such as pension funds, which use inflation swaps to hedge against inflation risks and ensure they can meet future payments; dealer banks that supply inflation swap contracts; and hedge funds engaging in arbitrage trading (Bahaj et al., 2023). The data for euro-area inflation-linked swaps are obtained from Bloomberg.

3.3. *Methodology*

We examine the impact of climate-related speeches on changes in inflation expectations.

Following Jordà (2005), we use local projections with four lags²⁶ to capture short-run dynamics

²⁵ Generally, financial market expectations are preferred over survey measures because they are directly tied to actual financial decisions. This means there is no risk of strategic response bias or discrepancy between what people say and what they truly believe. They are also available at different maturities from 1 to 10 years ahead (Hubert 2017). Related evidence in Coibion et al. (2020) indicates that, in low-inflation environments, conventional central bank communications often fail to effectively reach households and firms, so their inflation expectations tend to respond only weakly to monetary policy announcements.

²⁶ Information-criterion diagnostics (AIC, BIC, HQIC) indicate that a parsimonious specification with one to three lags is sufficient. Given the high frequency of the data and the large sample size, we adopt four lags to allow for flexible weekly adjustment dynamics. Results are similar under shorter lag lengths.

in the daily data in the analysis of the dynamic relationship between our variables of interest.²⁷

We estimate the following equation:

$$\Delta y_{t+h} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}, \quad (1)$$

where Δy_t : inflation-linked swaps (in differences), s_t : climate-related speeches (count: number of speeches per day, intensity: ratio of climate-related words to the total number of words in a speech per day, and dummies for different types of speakers), x_t : controls for other news and information released on the same day.

Actors with different institutional roles deliver the eurozone communication, therefore, we allow the effect of climate-related speeches to vary by speaker identity. In particular, we distinguish speeches by the ECB President (44 speeches), other members of the ECB Executive Board (78 speeches) and national central bank officials from the founding euro-area member states (342 speeches).²⁸ This distinction is motivated by differences in perceived informational advantage and agenda-setting power, as markets may treat some speakers as conveying more information about the likely direction, implementation, or internal consensus concerning climate policies.

We include several control variables: macroeconomic surprise variables, which are surprises in the releases of euro-area inflation (the Citi euro area inflation surprise index), unemployment (the euro area labor market surprise index), and business confidence (the euro area business confidence index), to control for macroeconomic data releases that might have an

²⁷ The local projection method can accommodate non-linearities in the transmission of monetary policy and is robust to model misspecification. The use of local projections as a bridge between applied macroeconomics and applied microeconomics is discussed in Jordà (2023).

²⁸ The speeches of the ECB President were not counted among those delivered by the members of the ECB Executive Board.

effect on inflation-swap rates and confound the observed relationships. The macroeconomic surprise data come from Bloomberg. Additionally, we include a dummy variable for all ECB Governing Council meetings (excluding those with major announcements of interest rate changes). We use surprise indices to capture unforeseen economic developments, whereas the ECB meeting variable accounts for systematic policy-driven influences on inflation expectations. Finally, we address potential spillovers from non-climate communication by controlling for non-climate speeches delivered on the same day.

We estimate all specifications with Newey–West robust standard errors to account for potential heteroskedasticity and autocorrelation in daily financial data. We conduct estimations in differences to reduce bias and improve efficiency, as inflation-swap rates display strong persistence and are consistent with near-unit-root behavior in levels, while first differences are stationary. The local projections provide us with impulse responses to evaluate the significance and persistence of the effects of central bank speeches on market inflation expectations. We estimate the regressions using daily data from January 1, 2008 (the earliest year with recorded speeches on climate change) through December 31, 2025.

One potential concern is *endogeneity*, which may arise if the content of climate-related speeches is systematically correlated with contemporaneous economic developments that are unrelated to climate, potentially biasing the estimates. To address this, we leverage the scheduling of ECB climate speeches, which are generally announced in advance (pre-scheduled speeches) and thus less likely to reflect immediate economic news or short-term fluctuations. This scheduling exogeneity allows us to treat the occurrence of climate speeches as largely independent of daily economic developments, reducing bias in our estimates. ECB staff typically prepare speeches in advance, especially for complex topics such as climate change, which

require a thorough, data-driven approach. Staff may adjust the content of speeches to reflect current developments, though such adjustments are generally focused on major events rather than everyday fluctuations.

4. Empirical Results

This section summarizes our key findings. We present the results from local projections of the impact of climate-related speeches on market-based inflation expectations, measured using inflation-linked swaps (ILS), while controlling for macroeconomic surprises, the ECB's monetary policy decisions, and non-climate speeches delivered on the same day.

4.1. Baseline Specification

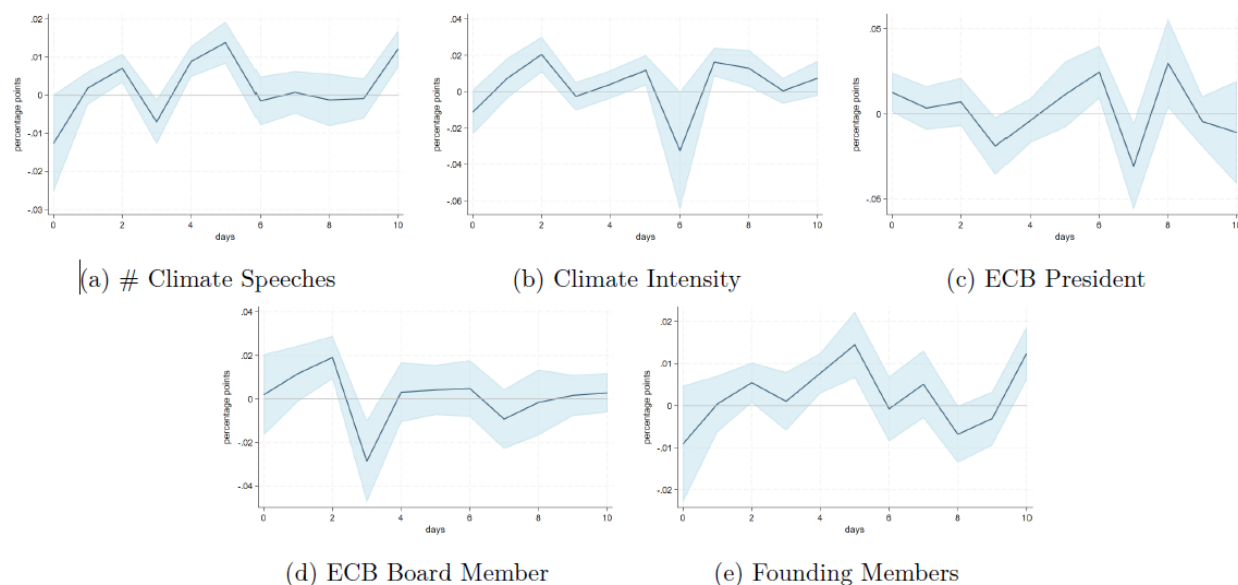


Figure 5: Impulse Responses of 1-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

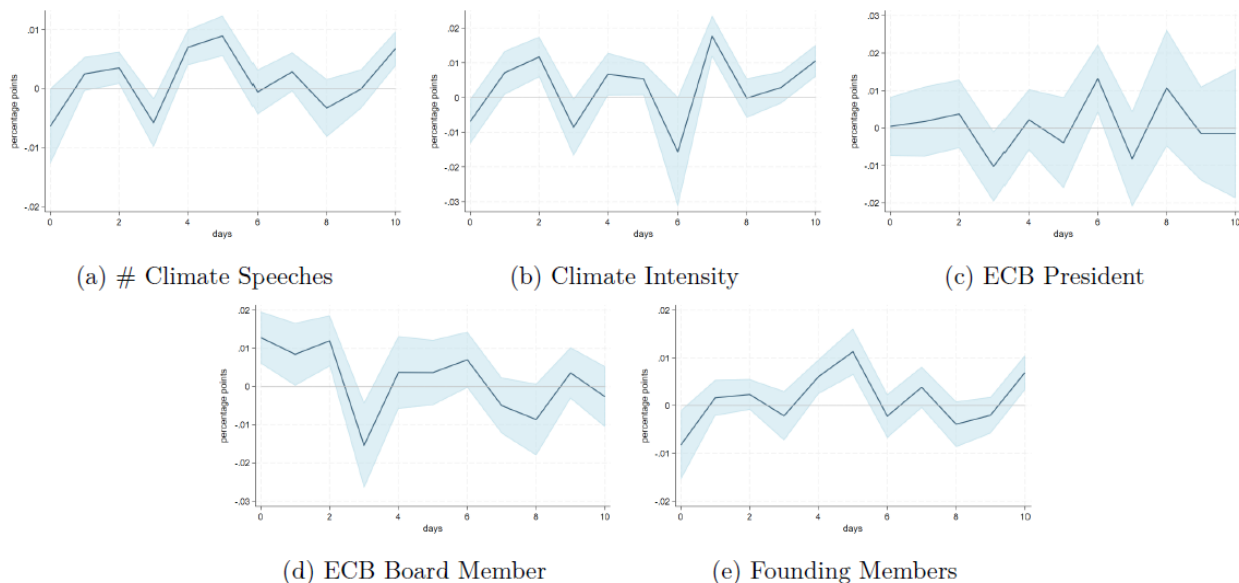


Figure 6: Impulse Responses of 2-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

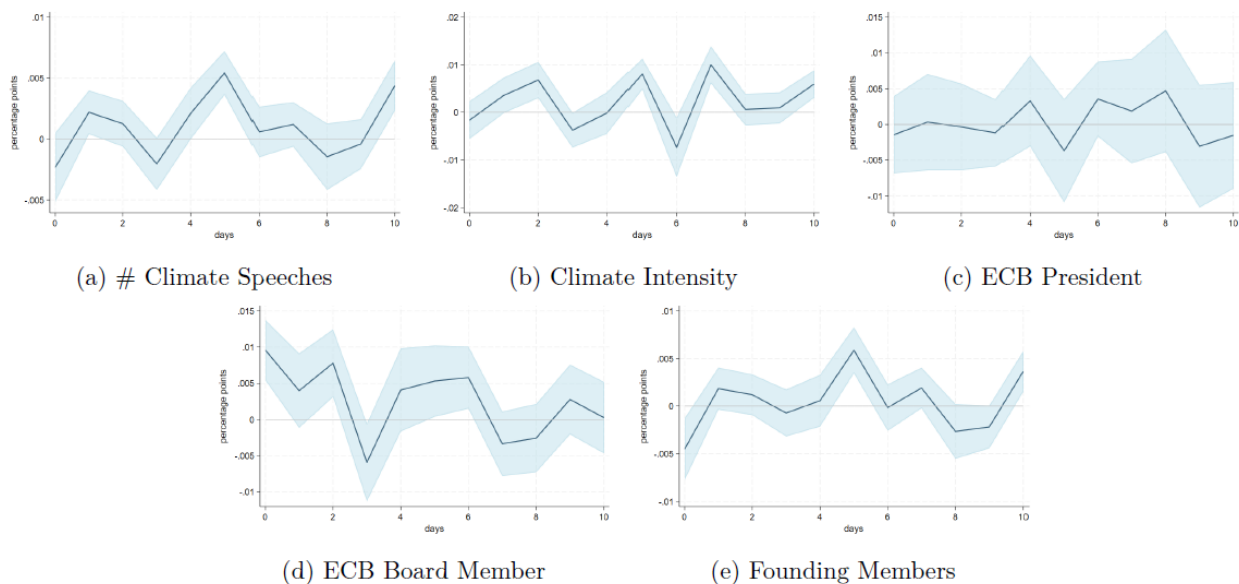


Figure 7: Impulse Responses of 5-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

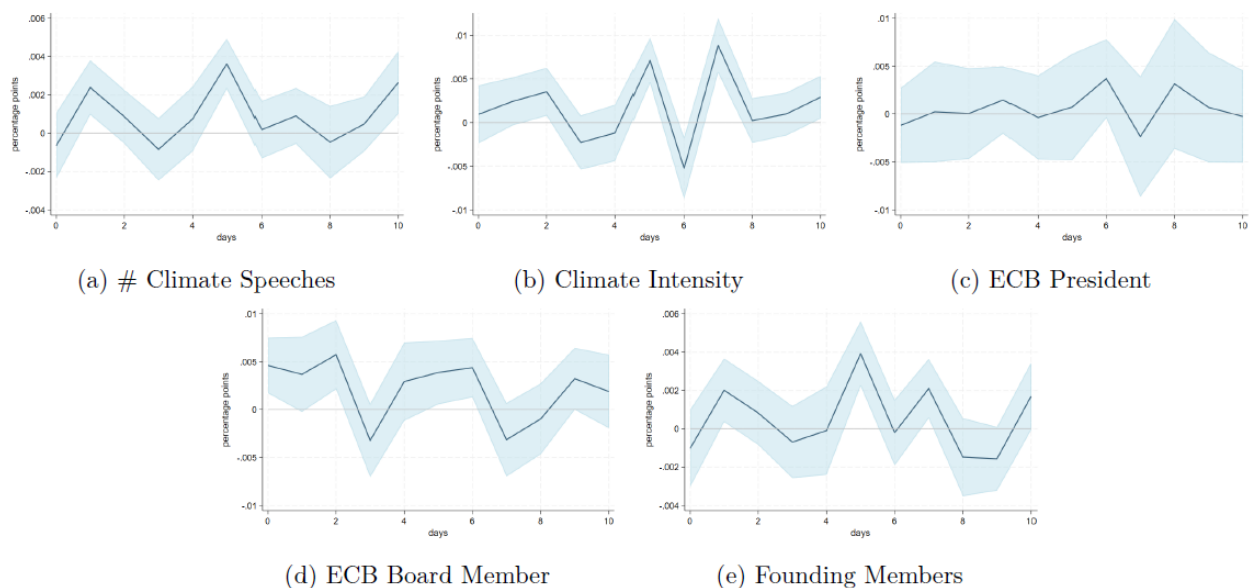


Figure 8: Impulse Responses of 10-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

Figures 5-8 illustrate the impulse responses of 1-year, 2-year, 5-year, and 10-year inflation-linked swaps to the frequency of climate-related speeches, the climate intensity, and speeches given by the President of the ECB, members of the ECB Executive Board, and representatives from central banks of the founding euro-area member countries. We present our baseline impulse response function estimates, obtained by estimating Equation (1).

At short maturities, 1- and 2-year inflation-linked swaps (Figures 5 and 6), the impulse responses are small and quite volatile in the first days, but stronger and a more visible positive response around day 4-5 after the speech. This indicates a slightly delayed response in inflation expectations to both the frequency (number) and intensity (ratio of climate-related words to total words) of climate-related speeches. In terms of the speakers' heterogeneity, speeches by central bankers from the founding member national central banks are associated with larger point estimates for short-term inflation expectations. In contrast, speeches by the President of the ECB

result in a weak and delayed market reaction on the sixth day, while speeches by ECB Executive Board members prove largely insignificant. The “up-and-down” movements suggest greater uncertainty and disagreement among market participants who interpret them as on the possibility of more restrictive monetary policy in the future to fight inflation, while others see the additional risks to price stability.

At medium- and long-term inflation-linked swaps (5-year and 10-year), depicted in Figures 7 and 8, both the frequency and intensity of climate-related speeches display a more systematic, modest upward shift in inflation expectations, typically peaking around the fourth and fifth lags. This indicates that markets price a mild upward revision to longer-run inflation risks. The impact of speeches by the ECB President remains largely insignificant, compared with the founding eurozone member officials and (at longer maturities) the ECB Board members. Appendix A presents the impulse response functions for how climate-related speeches affect inflation-linked swaps across various maturities (1, 2, 5, and 10 years), with 90% confidence intervals.

These results control for other (non-climate) speeches on the same day, which indicate that markets react specifically to climate content rather than to the fact that central banks are talking more on those days. Therefore, climate discourse adds a new layer of uncertainty on top of the usual monetary policy and macroeconomic commentary. The baseline empirical results thus provide evidence that climate-related central bank communication increases uncertainty about the monetary policy reaction function, leading to modest upward revisions in market inflation expectations. In sum, longer-horizon panels (5- and 10-year; Figures 7–8) display a more consistently positive pattern (the response spends more consecutive days on the positive side), while at short maturities (1- and 2-year; Figures 5–6) the responses are weaker and less

uniform over time (roughly day 4–5). This is consistent with a view that markets internalize climate-related communication as information about long-run inflation risks, while short maturities exhibit weaker and less stable reactions overall. Climate risks and transition policies are inherently long-term phenomena, while short maturities should react predominantly when speeches are interpreted as informative about short-term inflation dynamics (e.g., through energy-price channels).

While the relationship between speaker type and market responses is an empirical question, our estimates suggest that responses may vary across speaker types, though the evidence for speaker heterogeneity should be interpreted cautiously. Founding-member national central bank speeches display significant positive responses already at short maturities (Figures 5–6), while the Executive Board’s effects become more evident at longer maturities (Figures 7–8).²⁹ By contrast, the ECB President’s climate-related speeches display weaker and less precisely estimated effects. These findings suggest that different types of information related to the mandate versus implementation is delivered by different speakers. ECB President’s speeches usually provide high-level, consensus-oriented framing, aimed at explaining the ECB’s monetary policy. They shape expectations mainly by reinforcing the ECB’s legitimacy, with effects more likely to operate at longer horizons through expectation management and forward-guidance channels (Coibion et al., 2022). By contrast, Executive Board members and prominent euro-area national central banks (NCB) governors (e.g., Bundesbank, Banque de France) often convey more detailed, implementation-relevant information. Several Executive Board members have communicated frequently on climate-related issues, notably Isabel Schnabel, Frank Elderson (including during his tenure at De Nederlandsche Bank), and Luis de Guindos. Climate

²⁹ For example, Ambler and Rumler (2019) find that speeches delivered by ECB Executive Board members do not have a statistically significant impact on financial markets.

communication has also been prominent among national central bank leaders, such as Sabine Mauderer (a member of the Executive Board of the Bundesbank) and François Villeroy de Galhau (Governor of the Banque de France). Because national central bankers oversee monetary-policy operations and/or banking supervision, their speeches can signal operational follow-through (e.g., supervisory priorities, stress tests, disclosure requirements), prompting more immediate market updating at shorter maturities.

What accounts for investors' relatively modest and delayed response to climate-related speeches by central banks in the inflation swaps market across all time horizons? First, it is important to recognize that climate-related speeches, unlike those addressing monetary policy decisions (such as interest rates) or macroeconomic fundamentals (like inflation or unemployment), are *non-monetary* and therefore less anticipated. These speeches likely attract less attention than standard monetary policy press conferences, resulting in them being less frequently read and processed. Many were delivered at international central banking conferences over the weekend, as shown in Table 1. Consequently, investors would have to rely on subsequent communication through media, newspapers, or the official websites of the ECB or BIS to access information about the speech on the next business day. Assuming financial market participants received information about these climate-related speeches with some delay, our findings are consistent with those of Bahaj et al. (2023), who observed that responses in the UK inflation swap market "stabilize within one to three days and then persist over time." Similarly, we observe a market reaction within 2-3 days, followed by stabilization, as indicated by our confidence bands encompassing zero.

The short-lived, modest increase in inflation expectations in the days following a climate-related speech by central bankers suggests that financial markets pay attention to central bank

climate communication, interpreting it as a potential signal of potential changes in policy priorities or future inflation dynamics. The evidence that inflation expectations rise modestly in the days following a climate-related speech by central bankers, but this effect is short-lived, suggests that financial markets pay close attention to the communication of central bankers in the immediate aftermath, that is, showing some sensitivity but not overreacting.

In sum, financial markets' reaction is delayed due to the lower median salience of climate speeches, gradual diffusion through secondary reporting, and segmented market structure. Furthermore, because pointwise confidence intervals address significance at individual horizons, in robustness checks we evaluate whether responses are statistically detectable *jointly* over the post-speech window using Wald tests across horizons (Appendix B).³⁰

4.2. Cumulative Effects of Climate-Related Speeches

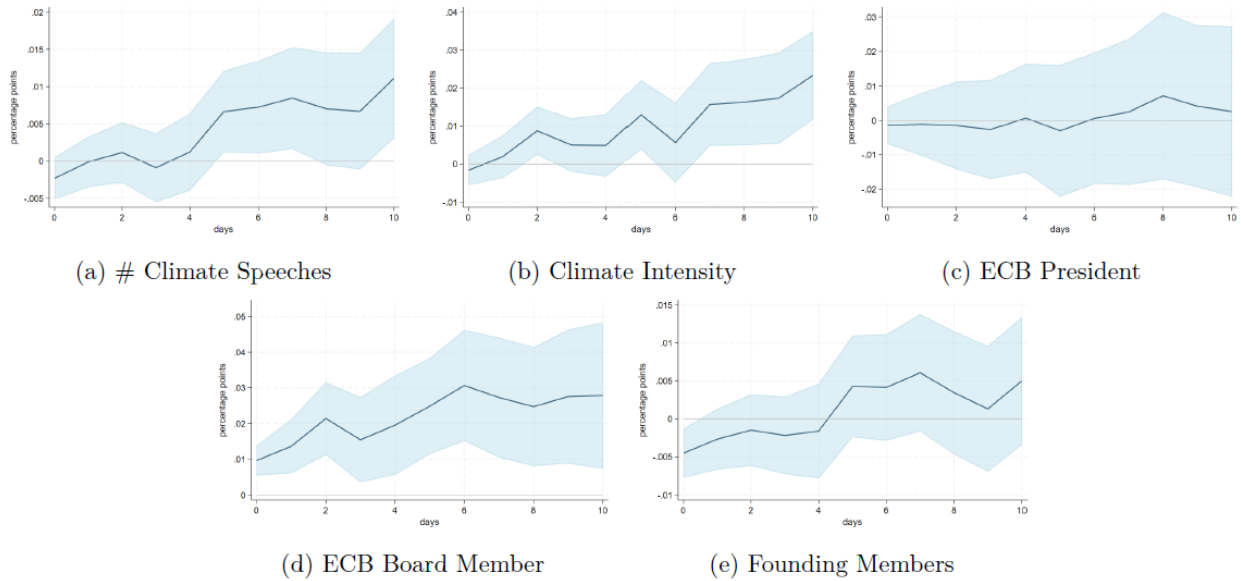


Figure 9: Cumulative Impulse Responses of 5-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation expectations, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

³⁰ We thank an anonymous referee for highlighting this point.

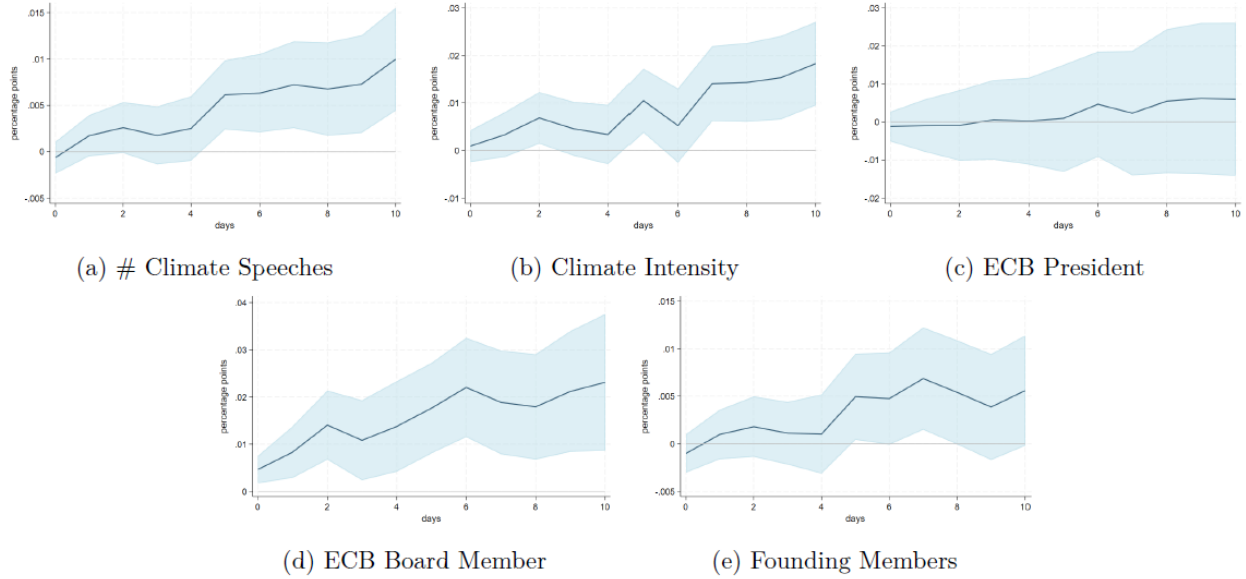


Figure 10: Cumulative Impulse Responses of 10-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-1} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation expectations, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

The cumulative impulse response functions confirm that climate speeches are associated with a persistent but modest upward shift in inflation expectations at longer horizons. We focus on longer maturities, as these reveal the most pronounced impact of climate speeches on inflation expectations.³¹ This finding is consistent with the long-horizon nature of climate risks and is theoretically grounded. For instance, Bansal et al. (2019) characterize climate change as a driver of long-run economic risk, implying that its effects are more readily observed through the pricing of long-duration assets. Similarly, Painter (2020) and Giglio et al. (2021) show that long-term assets exhibit greater sensitivity to climate-related shocks than short-term assets.

Figures 9 and 10 report the cumulative impulse responses of 5-year and 10-year inflation-linked swap rates to several measures of climate-related central bank communication. The

³¹ The results for shorter maturities are reported in Appendix B.

estimates display that the frequency and intensity of climate-related speeches are associated with a cumulative impact on 5-year and 10-year inflation-linked swap rates, and then flatten out at a small positive level. They suggest a persistent but bounded uncertainty effect, rather than de-anchoring inflation expectations. Cumulative effects are most pronounced when the speech comes from the ECB Executive Board Members, thus coming from speakers perceived as closer to policy implementation. In sum, a small positive plateau is more about a small repricing level shift, suggesting that markets may consider scenarios in which climate shocks or climate-related policies make it more difficult to keep inflation at the inflation target, rather than a runaway drift in long-term inflation expectations.

Intuitively, a one percent increase in the intensity of climate-related content in central bank speeches is associated with an estimated two basis point rise in both 5-year and 10-year inflation-linked swap rates. Interpreting magnitudes requires care because a “one percent” increase in salience is large relative to the distribution of climate intensity in our sample. As a result, around 2 basis point reaction should be viewed as an upper-tail comparison; typical changes in salience imply substantially smaller moves. This response is economically modest, consistent with the view that markets update but do not exhibit de-anchoring.³²

Several considerations help explain statistically detectable but economically modest baseline results. First, throughout most of this study's period, the eurozone experienced exceptionally low inflation (bordering on deflation) and inflation expectations remained well-anchored (Beechey, Johannsen, and Levin, 2011). Research shows that long-term inflation swaps

³² We would like to highlight the ways in which differences and cumulative impulse response functions (IRFs) complement each other. Differences IRFs capture daily changes that fluctuate and cross zero as the shock fades. Cumulative IRFs reflect the total effect over time. A small positive plateau in the Cumulative IRFs at longer horizons is consistent with Differences IRFs that taper off. In our context, anchored expectations mean that long-run inflation-linked swap rates remain stable and mostly unaffected by non-monetary communication. Therefore, a cumulative rise of about two basis points over two weeks reflects a modest response to climate-related communication rather than unanchoring.

generally exhibit minimal daily fluctuations and, if well-anchored, should show little reaction to new information (Picault, Pinter, and Renault 2022: 13).³³ When markets perceive the central bank as credible in meeting its inflation target, market-based inflation expectations tend to remain stable and show little sensitivity to monetary policy or macroeconomic news surprises, i.e., they are firmly anchored (see Garcia and Werner, 2018 for a review). Noticeable changes in inflation expectations are more likely to occur around policy innovations, such as announcements of unconventional monetary policies (Ambler and Rumler, 2019).³⁴

Second, our theoretical framework and empirical findings (in section 5.4.) point to offsetting opposing channels reflecting heterogeneity in narrative stance: some climate communication interpreted as increasing uncertainty about price stability mandate or highlighting inflationary climate effects, can raise inflation expectations, while other can neutralize those effects when it reaffirms the mandate. The average response can thus look small and short-lived, and markets revert to their prior expectations.

Finally, inflation-linked swap rates may reflect more than expected inflation because they can reflect time-varying inflation-risk compensation, liquidity premia, and related market frictions. Bahaj et al. (2023) emphasize that liquidity shocks and trading frictions are particularly important at short maturities, causing swap rates to deviate from underlying expected inflation and thereby attenuate estimated responses to climate speeches (biasing effects toward zero). Short-maturity estimates, in particular, may thus understate any response due to market frictions.

³³ Gürkaynak et al. (2010) and Beechey et al. (2011) show firmer anchoring of market-based long-term inflation expectations in the UK and euro area relative to the US.

³⁴ Neugebauer et al. (2024) argue that speeches are generally pre-scripted, which reduces their capacity to deliver unexpected information to markets. Therefore, the signaling content of speeches is typically weaker than that of, for example, interviews, which are more spontaneous, thus more likely to generate surprises.

4.3. Forward Volatility

We also assess the impact of climate-related speeches on the volatility of inflation-linked swaps, without assuming a specific direction of movement in inflation expectations. The rationale for this approach is as follows. Bahaj et al. (2023: 2) observe that the inflation swap market is highly fragmented into short-term and long-term segments, each controlled by different participants with divergent expectations. Pension funds are the primary buyers for long-term horizons (10 years or more), hedge funds actively trade in the short-term market (3 years or less), and dealer banks supply inflation swap contracts. Bahaj et al. (2023: 3) also point out that inflation expectations differ considerably between dealer banks and hedge funds. It is thus reasonable to assume that various agents in the inflation swap market interpret the implications of climate-related speeches differently, leading to increased volatility in inflation-linked swaps. Furthermore, a higher frequency and intensity of climate-related communication from central bankers is likely to amplify investor uncertainty, contributing to greater volatility in inflation expectations.

Figures 11 and 12 show the effect of climate-related speeches on the 5-day forward volatility of long-term inflation expectations, measured by 5-year and 10-year inflation-linked swaps. The estimates reveal that the frequency and content of climate-related speeches are associated with statistically significant positive volatility responses. Speaker heterogeneity flips relative to the level responses as the founding members of the eurozone and the ECB President's speeches generate the strongest and most lasting effects, while the speeches of the ECB Executive Board members do not exert a statistically significant effect. Uncertainty about the monetary policy reaction is reflected in detectably increased volatility of long-term inflation expectations and market fragmentation. The forward-volatility estimations provide the most

direct evidence consistent with the uncertainty prediction, because volatility responses are economically meaningful even when level effects are modest.

Overall, our findings indicate that climate communication is statistically detectable but does not generate evidence of unanchored inflation expectations; it does not trigger an immediate shock akin to a conventional monetary policy surprise, but a delayed effect. These patterns motivate the mechanism analysis that follows.

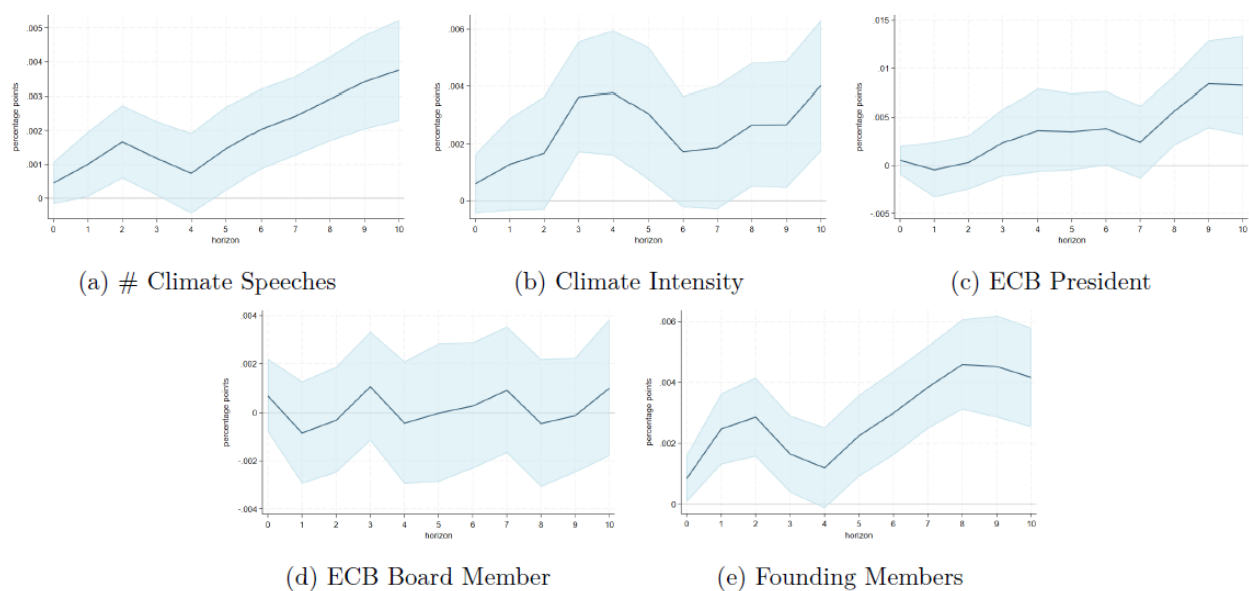


Figure 11: Impulse Responses of 5-year ILS (5-day forward volatility) to Selected Shocks (68% CI)

These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation-linked swap 5-day forward volatility, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate the 68 percent confidence intervals associated with local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

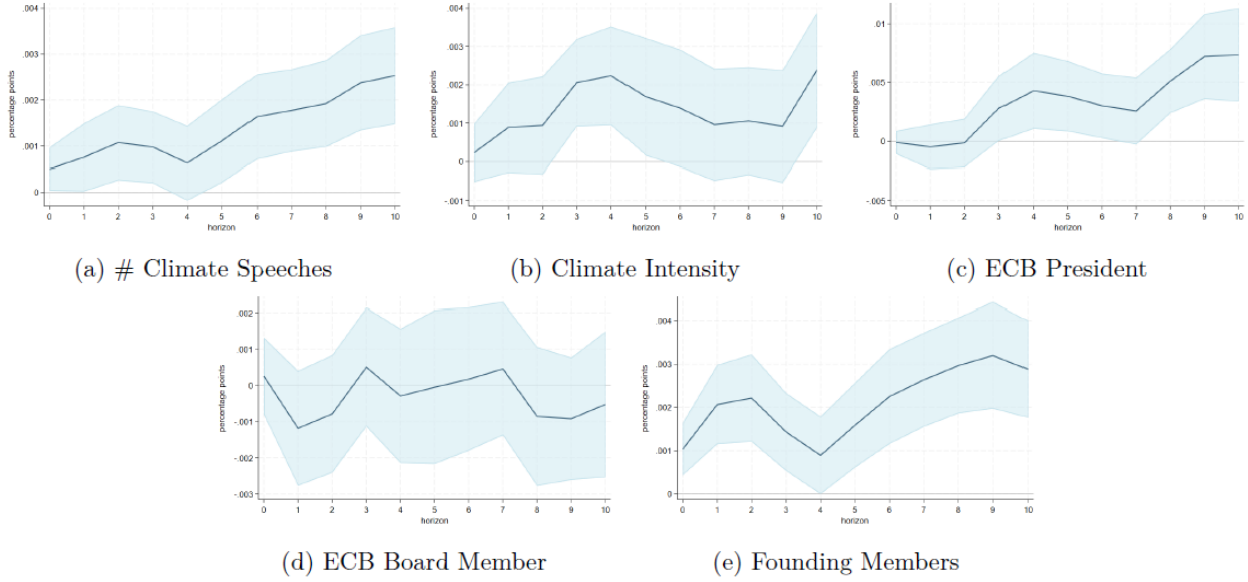


Figure 12: Impulse Responses of 10-year ILS (5-day forward volatility) to Selected Shocks (68% CI)
 These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation-linked swap 5-day forward volatility, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate the 68 percent confidence intervals associated with local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

4.4. Channel Decomposition

Baseline estimates show that climate speeches have a modest and delayed impact, with oscillating responses in first differences, more persistent cumulative effects at longer maturities, and higher forward volatility. Short-term heterogeneity may suggest a clash between mandate-drift and mandate-affirmation hypotheses; long-run persistence is consistent with either mandate-drift or new climate-inflation information; and volatility increase indicates the aggregate uncertainty prediction, but requires speech content decomposition. To disentangle these mechanisms, we decompose the climate speech sample by content stance.

4.4.1. Mandate-Drift Channel

Investors may believe that as the ECB expands its focus to include climate change, an area outside its traditional price stability mandate, it could be signaling a possible weakening of its

resolve to keep inflation in check. To capture this mandate-drift channel, we construct a mandate index that measures how central bankers discuss the institutional mandate in climate-related speeches. We first identify mandate-related sentences using the keyword anchor “mandate,” and then measure their sentiment using the Loughran–McDonald (LM) (2011) lexicon.³⁵ We define negative tone as the share of LM negative words relative to total words in these mandate sentences. A more negative tone around the mandate signals potential concerns about trade-offs or credibility risks associated with the ECB’s climate engagement. Our mandate-drift index, measuring negative tone in mandate-related sentences within climate speeches, is non-zero on 60 days over the sample period.

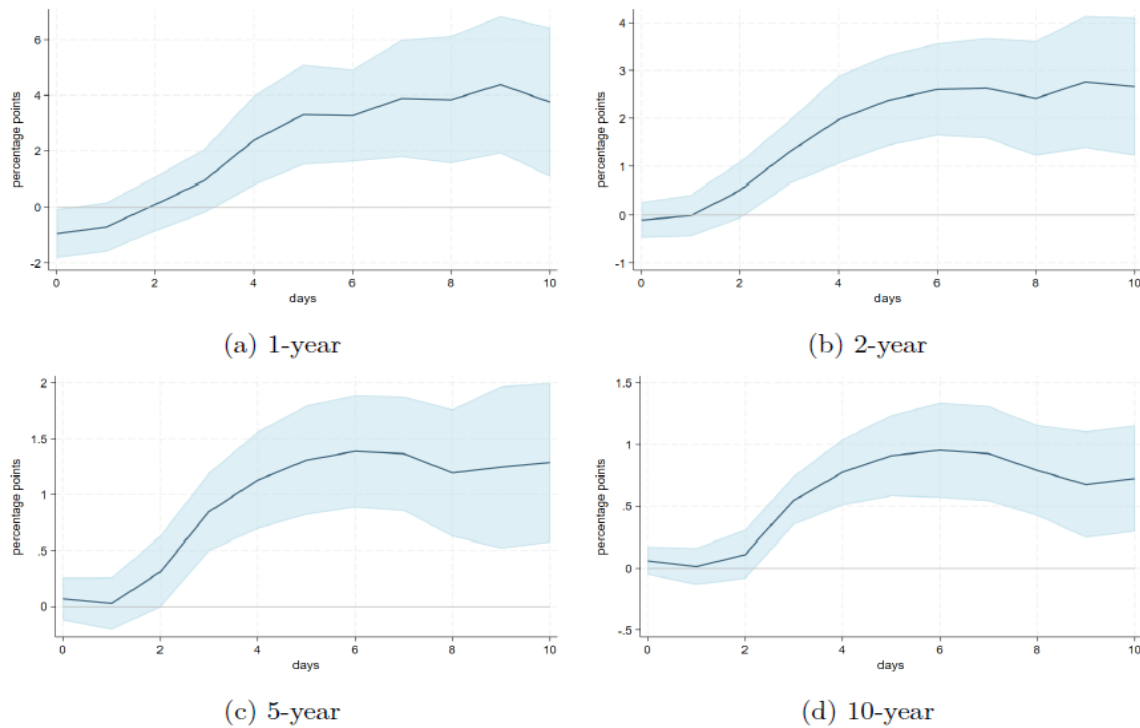


Figure 13: Cumulative Impulse Responses of ILS to the negative tone index related to mandate (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=l}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is a) 1-year ILS, b) 2-year ILS, c) 5-year ILS, d) 10-year ILS, and s is the mandate-drift index related to mandate. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

³⁵ <https://sraf.nd.edu/loughranmcdonald-master-dictionary/>

Our results indicate that when climate-related speeches contain mandate-related language with a more negative tone, cumulative impulse responses show upward pressure on inflation-linked swaps across maturities. These findings are consistent with our mandate-drift hypothesis suggesting that investors perceive this language as increasing uncertainty about future policy priorities and a potential weakening of the resolve to maintain price stability in favor of climate goals.

4.4.2. *Climate Inflation Information Channel*

To test the hypothesis that markets treat climate speeches as genuine inflation news, we construct an inflation-information index, which measures whether central bank communication signals upward or downward pressures on prices. The index is constructed by identifying sentences that reference inflation-related terms (“inflat,” and “price”) and counting additive modifiers (e.g., “rising,” “upward”) and subtractive modifiers (e.g., “easing,” “subdued”), following Granziera et al. (2025).³⁶ The index is defined as the normalized difference between additive and subtractive counts: $(add - sub)/(add + sub)$, with -1 assigned when only subtractive modifiers are detected, and 1 assigned when only additive modifiers are detected. Higher values indicate that climate change is associated with upward price pressures, whereas lower values suggest that climate is framed as disinflationary. The inflation-information index, capturing the net balance of inflationary (*add*) versus disinflationary (*sub*) modifiers in climate-proximal sentences, identifies 241 days containing either modifier type, of which 214 exhibit non-zero net

³⁶ Additive: "climb", "intensify", "jump", "elevat", "escalate", "expand", "foster", "height", "high", "increas", "persist", "pressure", "rise", "rising", "rose", "soar", "solid", "spik", "surg", "sustain", "strong", "strength", "upward", "up", "upside risk". Subtractive: "below", "collaps", "damp", "drop", "ease", "easing", "declin", "deteriorate", "diminish", "down", "low", "modest", "moderated", "muted", "reduction", "restrain", "set back", "slow", "soft", "subdued", "weak", "fall", "plummet", "retreat".

values (with 27 days netting to zero), which suggests that climate speeches often contain inflation framing rather than generic climate language. This content channel is more frequent in our sample than mandate-tradeoff rhetoric.

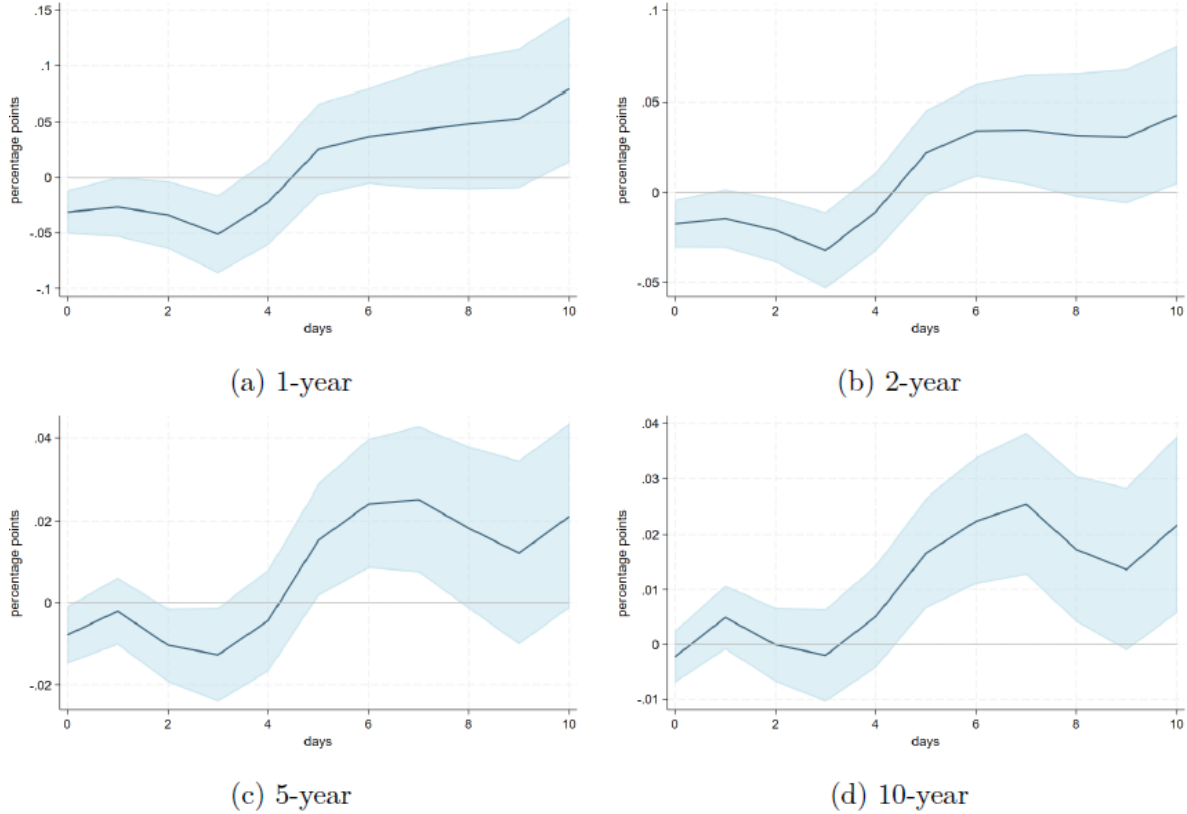


Figure 14: Cumulative Impulse Responses of ILS to the inflation-information index (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is a) 1-year ILS, b) 2-year ILS, c) 5-year ILS, d) 10-year ILS, and s is the inflation-information index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

The cumulative impulse responses show that speeches providing new information about upward price pressures are associated with higher inflation expectations, with the clearest and most consistent effects at medium-to-long maturities (5-10 years), consistent with markets perceiving these signals as persistent supply side pressures, rather than a short-lived cyclical shocks. The stronger response at longer maturities suggests that climate shocks (e.g., weather

disasters, energy and carbon prices) are perceived by investors as structural inflation pressures, thus central bankers' information of these risks is priced mainly into longer-term inflation swaps.

4.4.3. Mandate Affirming Channel

We test a mandate-affirming (credibility) channel by quantifying whether climate communication is perceived as more “hawkish” (tightening/price-stability affirming) or “dovish” (easing). The net stance index captures the overall hawkish or dovish orientation of monetary policy language. It is computed by identifying sentences discussing interest rates (“rate”) and counting hawkish terms (e.g., “tighten,” “raise,” etc.) and dovish terms (e.g., “ease,” “accommodative”, etc.) from Tobback et al. (2017), adjusting for local negations that reverse their meaning.³⁷ The index is defined as $(H-D)/(H+D)$, where H and D denote hawkish and dovish counts, respectively, and equals -1 when only dovish terms appear, and 1 when only hawkish terms appear. Thus, higher values indicate a more contractionary policy signal, while lower values indicate an easing signal. We construct this measure in line with existing scholarship (e.g., Malmendier et al., 2021; Apel et al., 2022) to capture whether climate-related communication coincides with perceived shifts in the central bank’s policy. The mandate-affirming index, representing the net balance of hawkish minus dovish monetary policy terms in climate-proximal sentences, identifies 173 days containing either term type, of which 157 exhibit non-zero net values and 16 zero values.

³⁷ Hawk: “raise”, “increase”, “go up”, “tighten”, “head up”, “hike”, “move up”, “put up”, “rise”, “lift”, “boost”, “bump up”, “augment”, “higher”, “climb”, “hawkish”, “tight”. Dove: “cut”, “decrease”, “go down”, “ease”, “head down”, “slash”, “move down”, “put down”, “reduce”, “loosen”, “slice”, “shave”, “trim”, “lower”, “drop”, “dovish”, “loose”, “accommodation”, “accommodative”. Negation words: “not”, “unlikely to”, “no reason to”.

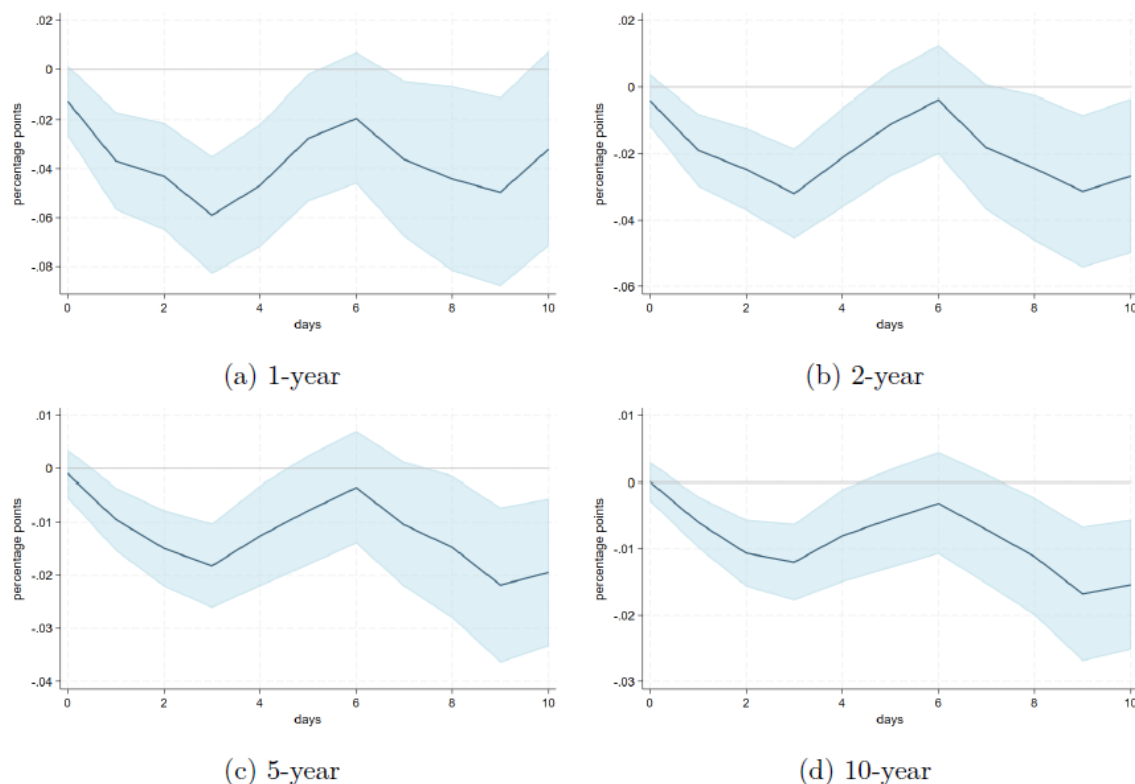


Figure 15: Cumulative Impulse Responses of ILS to the net stance index (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=l}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is a) 1-year ILS, b) 2-year ILS, c) 5-year ILS, d) 10-year ILS, and s is the mandate-affirming index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

Cumulative impulse responses show that climate speeches framing climate action as consistent with the primary mandate of price stability (through a hawkish stance to anchor expectations) produce lower or stable inflation expectations. These hawkish, mandate-consistent climate speeches neutralize upward pressure on inflation expectations from the previous mandate-drift and climate-inflation channels, showing that credibility signals can offset climate-induced uncertainty and prevent de-anchoring. Inflation-linked swaps include the inflation risk premium (Bahaj et al. 2023); thus, a positive response can reflect higher inflation risk compensation under greater uncertainty, as our uncertainty mechanism would predict, even if mean expectations move modestly.

4.4.4. Financial Stability

Financial stability is a supporting function in the ECB's institutional framework; climate-related speeches may thus influence inflation expectations if markets expect financial-stability interventions that ultimately require monetary accommodation. Furthermore, climate speeches might just signal financial risk management, not mandate drift.³⁸

Climate change threatens financial stability: stranded assets and the repricing of climate exposures can generate significant losses for the financial system, and data limitations can hinder investors' ability to price these risks accurately (Bolton and Kacperczyk, 2021). It may also give rise to "green swan" events, which are extreme, unpredictable shocks that destabilize financial markets. Dikau and Volz (2021: 16) caution that climate-related risks may precipitate a "climate Minsky moment," in which a sudden reassessment of physical or transition risks causes a sharp decline in asset values and strains banks, insurers, and other financial institutions. In such a scenario, markets may expect central banks to act as "climate rescuers of last resort," providing emergency liquidity to financial institutions affected by climate-related losses. (Bolton et al., 2020). But if investors start to view this emergency liquidity support as a recurring response, they may build in bailout expectations, which can encourage risk-taking and contribute to credit expansions and asset-price booms (e.g., in stock or real estate markets). Since stabilizing the financial system often involves liquidity injections or monetary easing, these beliefs can lead markets to anticipate future policy accommodation, putting upward pressure on inflation expectations.

We separate the mandate from financial stability-focused speeches to capture how climate communication relates to systemic risk considerations. We use the financial stability

³⁸ We are grateful to the anonymous referee for raising this concern.

dictionary from Arseneau et al. (2022) to identify sentences referencing financial stability (e.g., "financial stability", "systemic risk", "macroprudential policy", "resolution regime", "macroprudential instrument", "countercyclical capital buffer", "macroprudential perspective"). In the next step, we calculate an “uncertainty” sentiment using the Loughran–McDonald lexicon, following the same method as for the negative mandate tone index. A higher uncertainty sentiment indicates that climate issues are framed as potential threats to financial stability (such as asset repricing, or financial system vulnerabilities). These measures allow us to assess whether markets interpret climate communication as a proxy for financial stability shocks (e.g., green swan events, stranded assets) rather than mandate uncertainty. The financial stability uncertainty index is defined as Loughran-McDonald uncertainty words divided by total words (nonzero on 171 days in our sample).

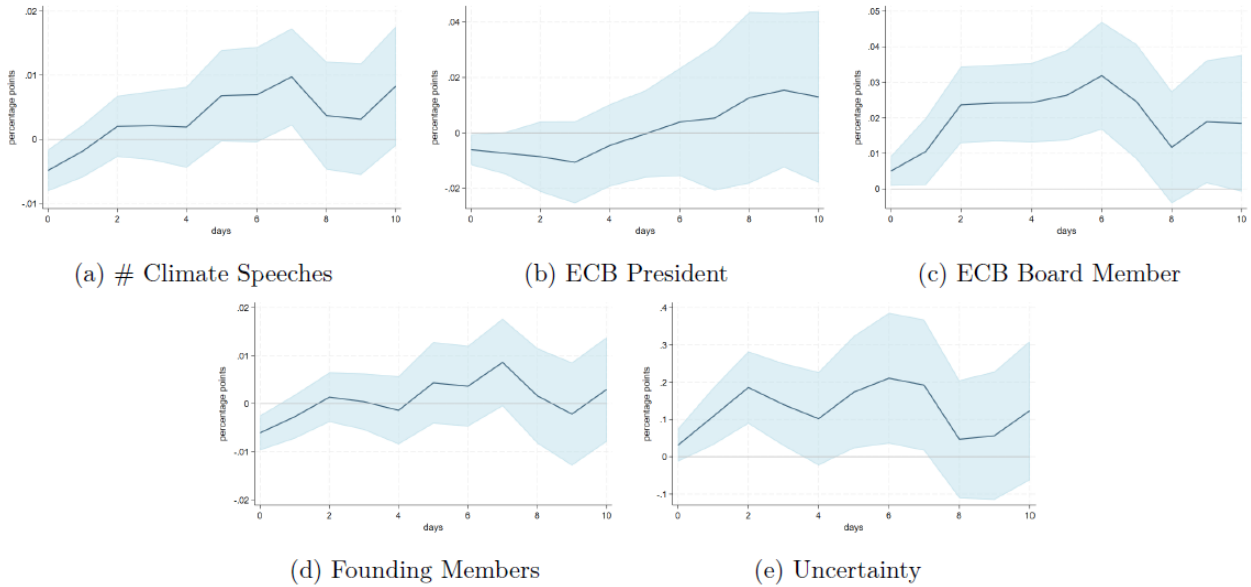


Figure 16: Cumulative Impulse Responses of 5-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-1} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation expectations, and s is a) # of climate speeches, b) speeches delivered by the ECB President, c) by an ECB Board Member, d) by a representative of founding member countries, and e) the financial stability uncertainty index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

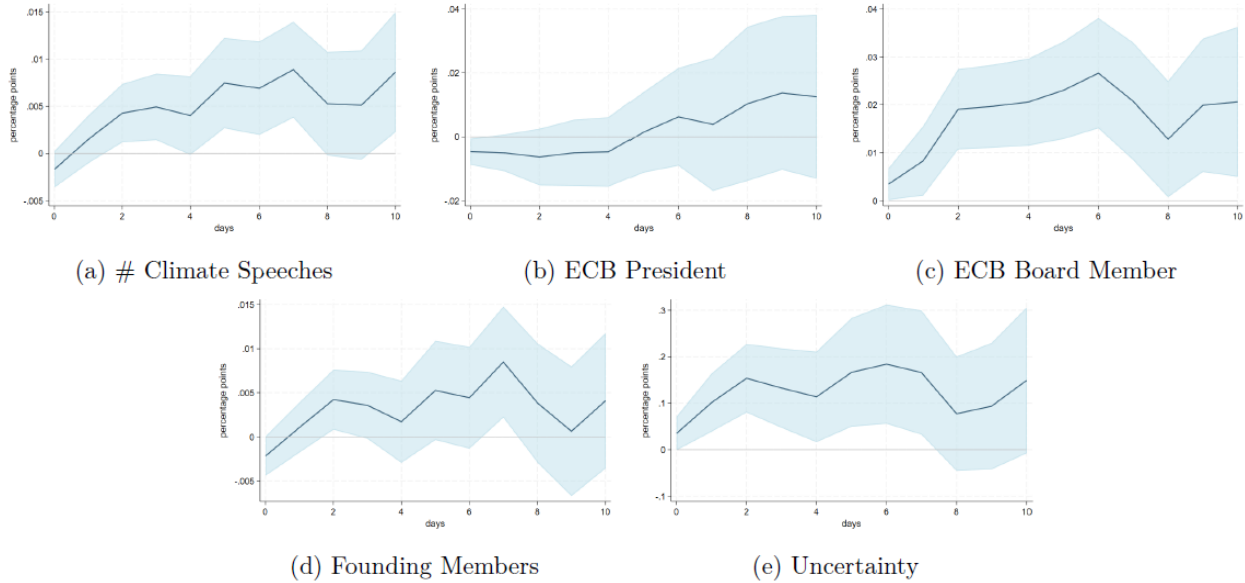


Figure 17: Cumulative Impulse Responses of 10-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation expectations, and s is a) # of climate speeches, b) speeches delivered by the ECB President, c) by an ECB Board Member, d) by a representative of founding member countries, and e) the financial stability uncertainty index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

Figures 16-17 show that financial stability climate communication is associated with an upward revision in inflation expectations at medium to long maturities (5-10 year inflation-linked swaps). This is consistent with market participants interpreting climate-related financial stability concerns as increasing the likelihood of future monetary accommodations, such as liquidity provision during climate-induced financial disruptions (e.g., “green swan” events). The effects are often most pronounced at medium-to-long maturities, reflecting the long-term nature of green swan risks and the potential for persistent supply-side shocks to the financial system. The findings also indicate that tone matters, as the responses are strongest when financial stability discussion involves higher uncertainty sentiment, suggesting that it is not only mention of financial stability that influences market expectations but also communication that highlights vulnerability and hard-to-price risks. Finally, the effect is strongest for speeches delivered by

members of the ECB Executive Board, consistent with markets attributing greater informational content and systematic risk management. Short-term maturities (1-2 years) show more volatile responses (Appendix C).

5. Robustness and Extensions

We confirm robustness across VAR models, the ECB pre-meeting "quiet period", inflation-linked bonds, joint Wald tests, and extended controls. We relegate some of these extensions to Appendices, while others are available upon request from the authors.

To address concerns about speech timing near policy events, we include a dummy variable for speeches delivered within seven days of a monetary policy meeting during the so-called “quiet period,” when the ECB's Governing Council members are expected to refrain from making comments that could influence expectations about forthcoming monetary policy decisions. Appendix D shows that when we control climate speeches occurring within a 7-day window of ECB meetings, their effects on inflation expectations are qualitatively and quantitatively consistent with our baseline specification. This estimation suggests that the timing of climate speeches in relation to policy meetings and announcements has a minor impact on shaping market expectations.

Vector autoregressive models offer an alternative approach to estimating impulse response functions to local projections (Plagborg-Moller and Wolf, 2021). Local projections have some advantages over VAR models: they are straightforward to estimate and make inferences from, impose fewer constraints on the data-generating process, are less sensitive to misspecification than VARs, and are more flexible in handling state-dependent and non-linear

specifications. Appendix E demonstrates that the results obtained from VAR estimations align closely with those from local projections.

To further engage with the limitations of inflation-linked swaps, particularly that short-term maturities may be unreliable proxies for expected inflation due to dominant liquidity frictions (Bahaj et al. 2023), we conduct robustness checks using inflation-linked bonds, following Bekaert and Wang (2010). These instruments exhibit lower liquidity premia and reduced arbitrage frictions at short horizons, providing a cleaner measure of inflation compensation. The results (Appendix F) confirm the baseline findings that climate intensity generates modest upward pressure, which is most systematic at longer maturities, while short-term responses remain modest.

Finally, we include additional or alternative control variables.³⁹ We extend our regression specifications by including dummies for days when there was a change in the ECB policy rate and for days when the ECB lowered interest rates. We also incorporated controls for financial uncertainty using the VIX, officially called the Chicago Board Options Exchange Volatility Index, which measures the level of volatility that professional investors expect for the S&P 500 index over the next 30 days. This is known as "implied volatility" because the VIX is based on the options market, where traders speculate on the future performance of various securities and market indices. We assess the robustness of our results to uncertainty in European financial markets by incorporating the VSTOXX and EURO STOXX 50 indices. We also control for oil price shocks by including the price of one-year-ahead future contracts on West Texas Intermediate Crude oil from Bloomberg. As an additional control for global uncertainty, we

³⁹ The results are available from the authors upon request.

utilized the Monetary Policy Uncertainty Index (Husted et al., 2020), constructed by identifying monetary policy uncertainty keywords in US newspaper archives. The results remain consistent with the baseline findings.

5.1. Joint Wald Tests across Horizons

Appendix B reports the results of estimations using joint (multi-horizon) Wald tests to assess whether climate-related communication generates statistically detectable multi-day responses in inflation-linked swaps, complementing the impulse-response figures. We compute standard errors using Newey–West estimators with a horizon-specific bandwidth to account for the serial correlation induced by overlapping forecast horizons. To assess whether the response is statistically different from zero over horizons $h = 1, \dots, 10$, we test the joint null hypothesis that the impulse-response coefficients are equal to zero across these horizons using a Wald statistic based on the estimated cross-horizon covariance matrix of the coefficients.⁴⁰

Table B.1 shows that climate intensity is jointly significant at all maturities, meaning that more climate-focused communication is systematically associated with changes in market-based inflation expectations. Speech frequency is jointly significant at medium term maturities (2- and 5-year) and less robust at longer maturities. The Joint Wald tests also alter our interpretation of speaker heterogeneity: speaker type effects are generally weak under joint inference, with significance mainly at the short-term maturities for founding-member speakers and limited evidence for the other speaker categories. We therefore treat the baseline “who speaks” findings as suggestive and highlight that the market response is primarily content-driven.

⁴⁰ See Jordà (2009) for joint/simultaneous inference for impulse responses. Inoue and Kilian (2016) provide related joint inference procedures for structural impulse responses.

6. Conclusion

This study explores how climate-related communication affects market inflation expectations. Drawing on a novel dataset of climate-related speeches by the ECB and euro-area national central bank officials from 2008-2025, local projections estimates show both the frequency and the intensity of climate-related central bank speeches have a statistically detectable but economically modest effect on inflation-linked swap rates, with impacts appearing with a delay (and most clearly at longer maturities). We interpret these aggregate responses as operating primarily through uncertainty about mission creep and future policy trade-offs. We further decompose the overall effect across three content-based mechanisms. First, a mandate drift channel, where negative tone in mandate-related sentences is associated with higher inflation expectations, consistent with market concerns that climate objectives could conflict with price stability. Second, climate-inflation information highlighting inflationary consequences of climate risks or the green transition, associated with increased inflation expectations. Third, a mandate-affirming channel, climate speeches that reasserts price stability, often with a hawkish monetary policy framing, neutralize upward pressure on inflation expectations. In addition, separating out a financial stability framing (distinct from mandate concerns) yields similarly positive responses, with effects most pronounced at medium-to-long term maturities.

These impulse-response functions also provide some evidence that the effects of these speeches can vary depending on who delivers them and when they are made relative to policy decisions. Speeches by officials from the euro area's founding member countries and by the ECB Board members are associated with a small but statistically significant increase in measured inflation expectations. However, once we account for joint inference and dependence across overlapping horizons, we find that markets respond mainly to content salience and not who

speaks (institutional rank). A broad set of robustness checks, including controls for macroeconomic surprises, central bank decisions, and non-climate speeches, the use of inflation-linked bonds as an alternative dependent variable, and specifications based on vector autoregressions, confirm that markets respond to climate-related content in central bank communication.

In sum, we document modest but statistically detectable effects of non-monetary communication, which is nontrivial, given that market-based inflation expectations are well anchored in the euro area. These results demonstrate that financial markets are attentive to climate-related central bank communication, but also show that such communication does not lead to persistent or destabilizing changes in inflation expectations, which is consistent with credible inflation targeting. The findings may suggest that climate rhetoric requires markets to parse complex narratives regarding future policy trade-offs.

More broadly, our findings contribute to the ongoing policy debate regarding the role of central banks in addressing climate change and the risk of "mission creep." We provide evidence that climate-focused speeches by euro area central bankers do not generate detectable de-anchoring effects in inflation expectations, despite concerns raised in the mandate-expansion debate. We also show that climate narratives focusing on mandate-tradeoff and inflation information push expected inflation expectations, while mandate-affirming speeches stabilize them. Thus, how the central bank talks about climate change, the precise framing of climate speeches, may be more important than whether it talks about it. The temporary increase in inflation expectations following central bank climate speeches may suggest that markets can distinguish between acknowledging climate risks and actual changes in monetary policy. Central banks can address climate change in their public communication without jeopardizing their main

objective if they maintain a transparent and credible commitment to price stability. Future work can explore the different types of climate rhetoric, analyze responses from varying market segments, or extend this research to other central banks.

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APPENDIX A: Estimated effects of climate-related speeches on inflation-linked swaps, 90 % confidence intervals.

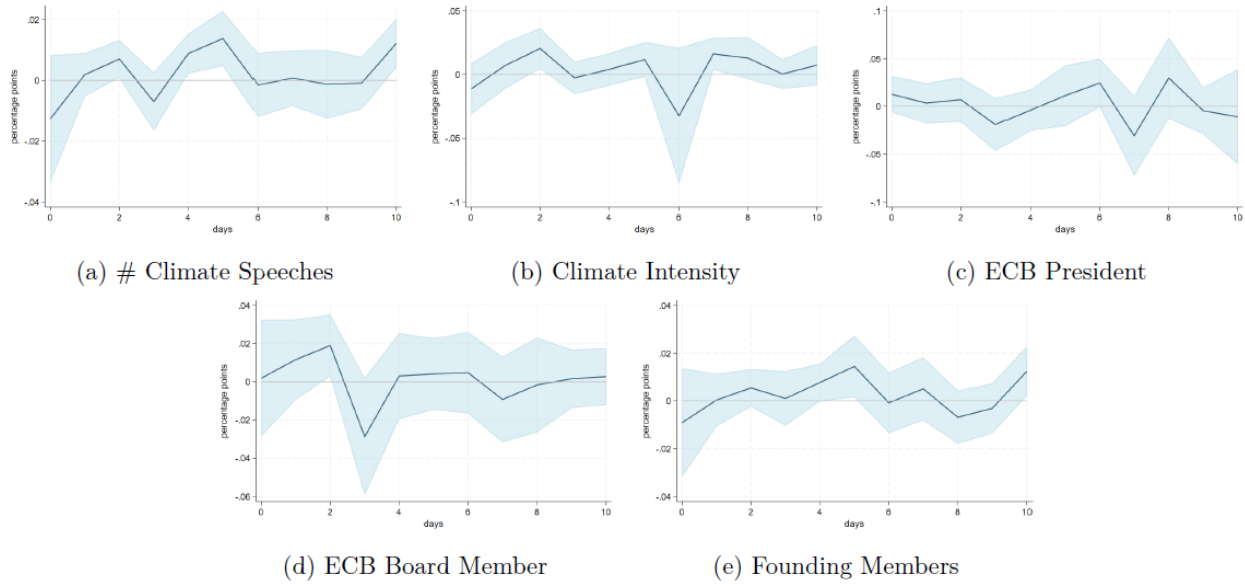


Figure A.1: Impulse Responses of 1-year ILS to Selected Shocks (90% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

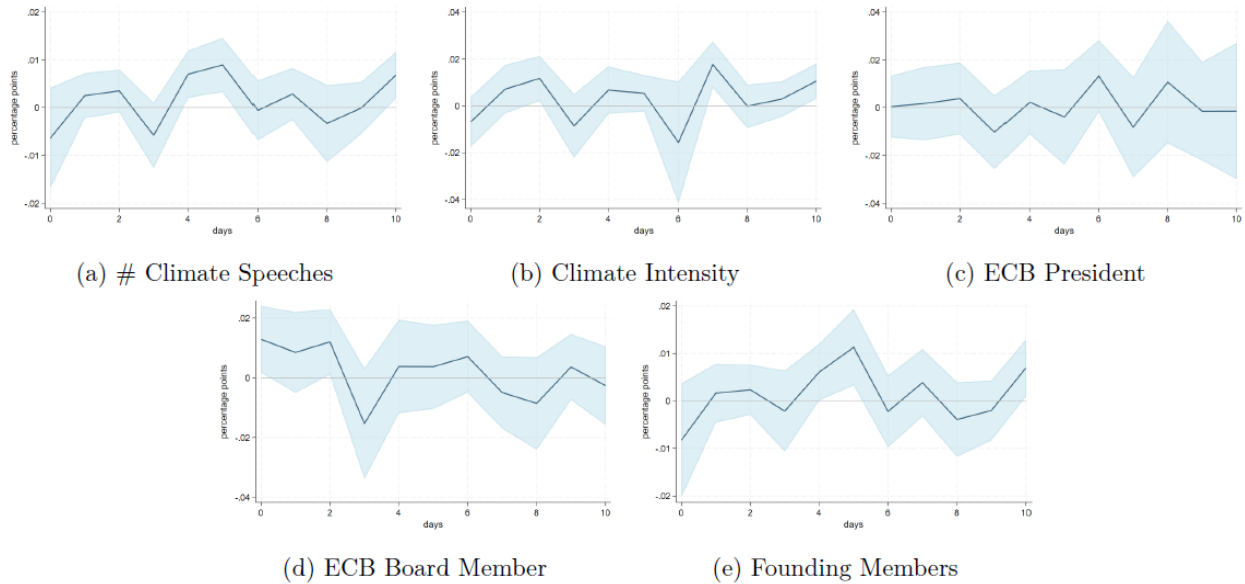


Figure A.2: Impulse Responses of 2-year ILS to Selected Shocks (90% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

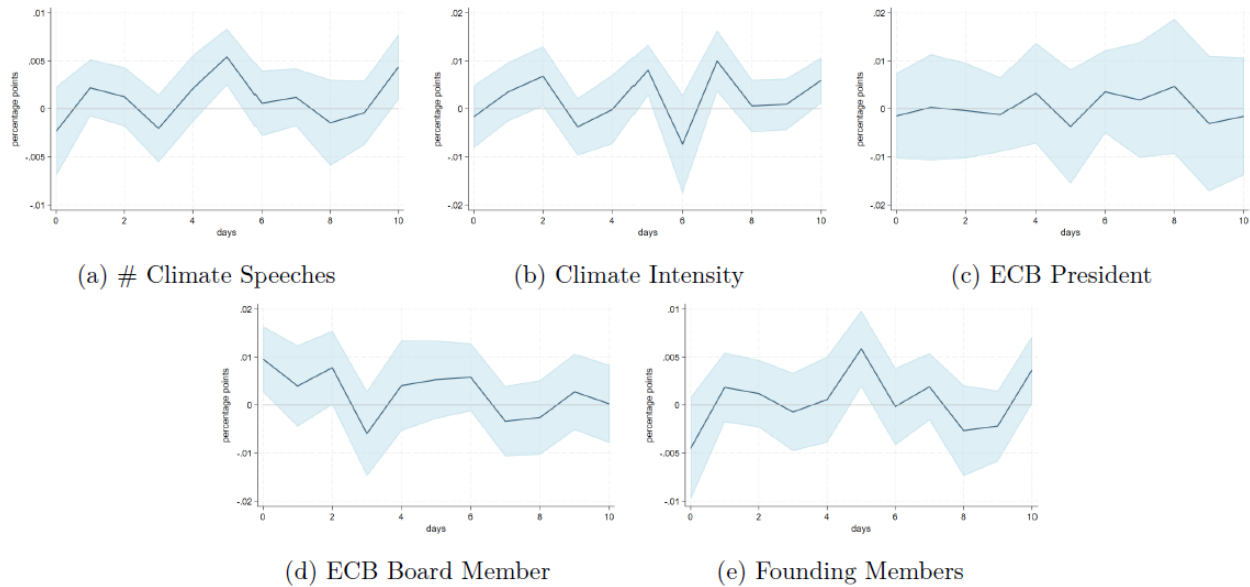


Figure A.3: Impulse Responses of 5-year ILS to Selected Shocks (90% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

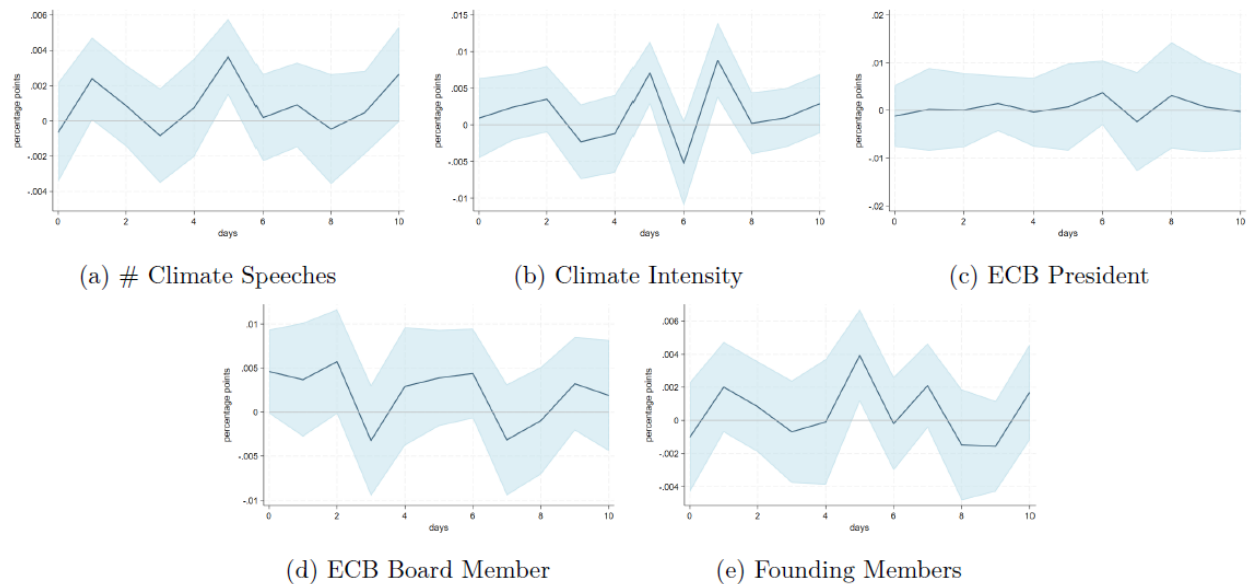


Figure A.4: Impulse Responses of 10-year ILS to Selected Shocks (90% CI)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

APPENDIX B: Estimated impacts of climate-related speeches on inflation-linked swaps and Wald test

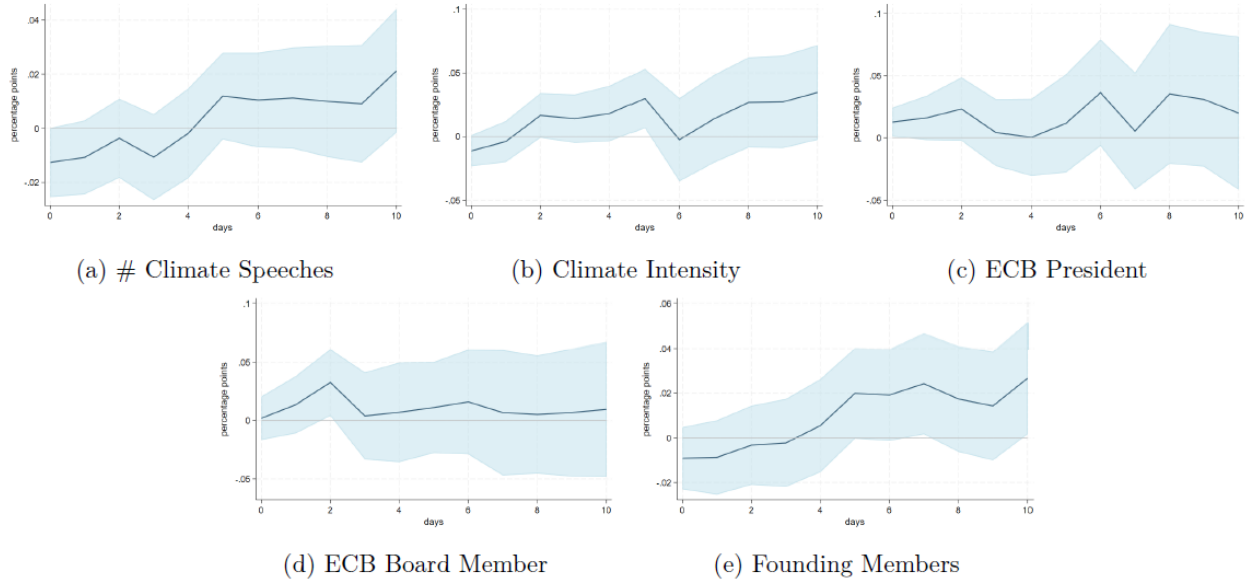


Figure B.1: Cumulative Impulse Responses of 1-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=l}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation expectations, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

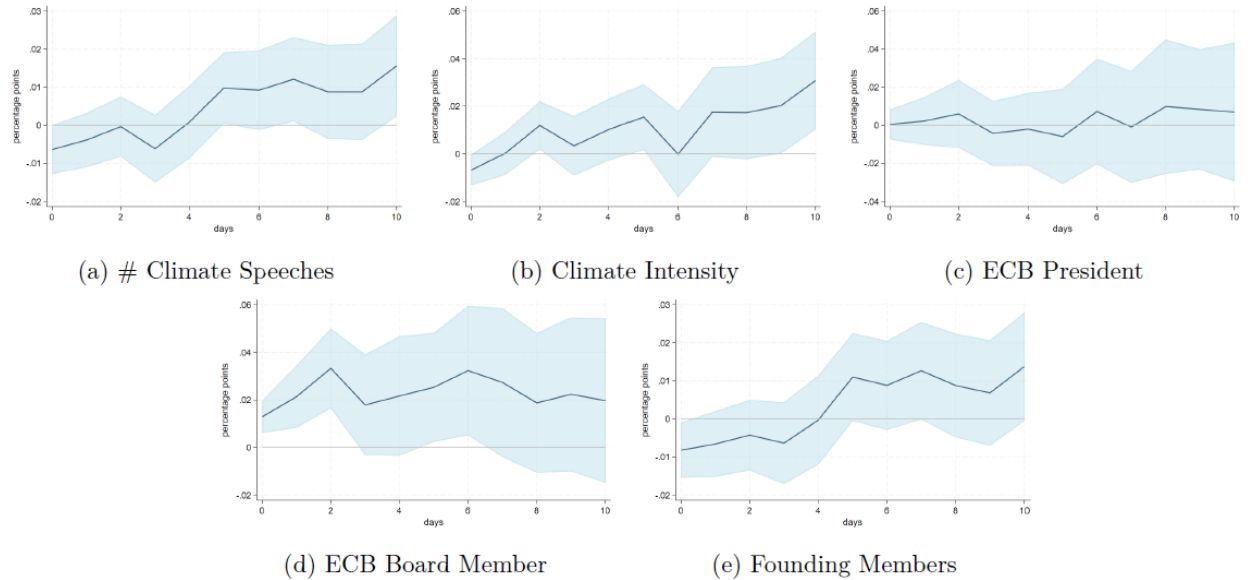


Figure B.2: Cumulative Impulse Responses of 2-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-l} = \alpha_h + \sum_{m=l}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation expectations, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

Table B.1: Wald test

Panel A: Differences IRFs				
	1-year	2-year	5-year	10-year
# Climate Speeches	0.053*	0.039**	0.036**	0.104
Climate Intensity	0.067*	0.011**	0.007***	0.007***
ECB President	0.354	0.066*	0.603	0.546
ECB Executive Board Member	0.116	0.257	0.198	0.137
Founding Members	0.14	0.049**	0.242	0.458
Panel B: Cumulative IRFs				
	1-year	2-year	5-year	10-year
# Climate Speeches	0.020**	0.037**	0.101	0.276
Climate Intensity	0.114	0.028**	0.018**	0.017**
ECB President	0.414	0.091*	0.743	0.581
ECB Executive Board Member	0.068*	0.273	0.212	0.171
Founding Members	0.086*	0.043**	0.269	0.627

Panel A reports the p-values of Wald tests for the Differences IRFs shown in Figures 5-8. Panel B reports p-values of Wald tests for the Cumulative IRFs shown in Figures 9-10 and B.1-2. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

The shaded areas in the Figures reported in the paper represent 68% confidence intervals (following Bahaj et al., 2023) to summarize the shape and timing of responses. Pointwise intervals test $H_0: \beta_h = 0$ at a single horizon h . Instead, to assess whether the response is statistically different from zero over the full post-speech window, Table B.1 reports a joint Wald test of $H_0: \beta_1 = \dots = \beta_{10} = 0$ using the estimated cross-horizon covariance matrix. When we

refer to statistically detectable multi-day responses, we mean rejection of this joint null; point estimates are used to describe when responses peak (around day 4–5), not to claim persistence. We compute standard errors using Newey–West estimators with a horizon-specific bandwidth to account for the serial correlation induced by overlapping forecast horizons. To assess whether the response is statistically different from zero over horizons $h = 1, \dots, 10$, we test the joint null hypothesis that the impulse-response coefficients are equal to zero across these horizons using a Wald statistic based on the estimated cross-horizon covariance matrix of the coefficients.⁴¹

⁴¹ See Jordà (2009) for joint/simultaneous inference for impulse responses. Inoue and Kilian (2016) provide related joint inference procedures for structural impulse responses.

APPENDIX C: Estimated impacts of financial stability speeches on inflation-linked swaps

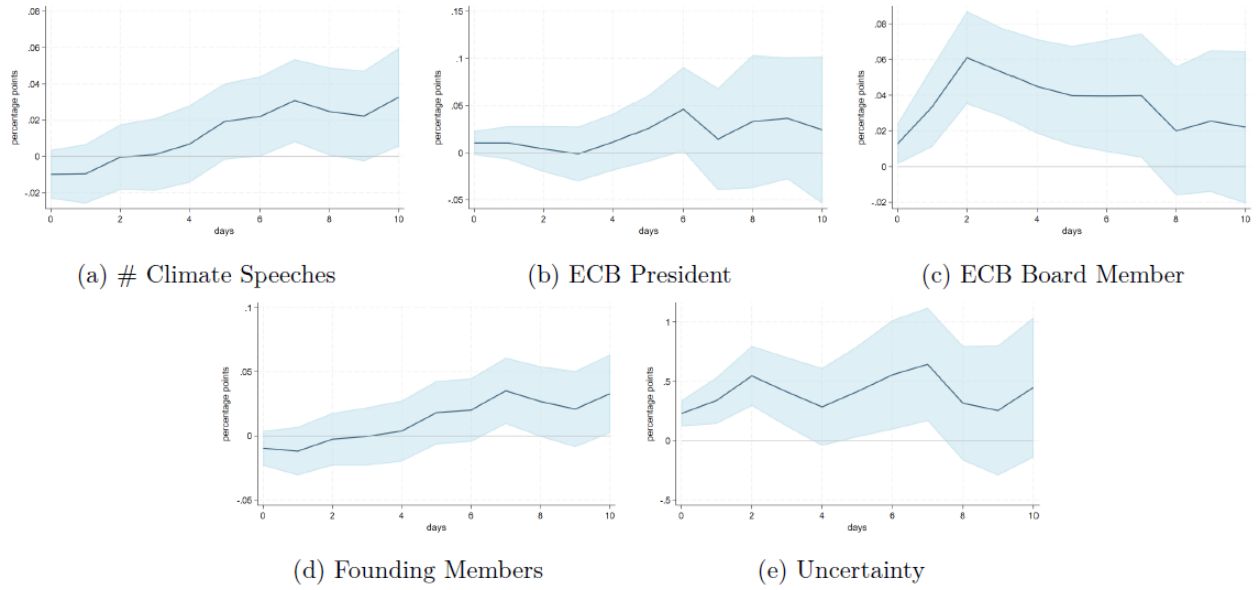


Figure C.1: Cumulative Impulse Responses of 1-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-1} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation expectations, and s is a) # of climate speeches, b) speeches delivered by the ECB President, c) by an ECB Board Member, d) by a representative of founding member countries, and e) the financial stability uncertainty index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

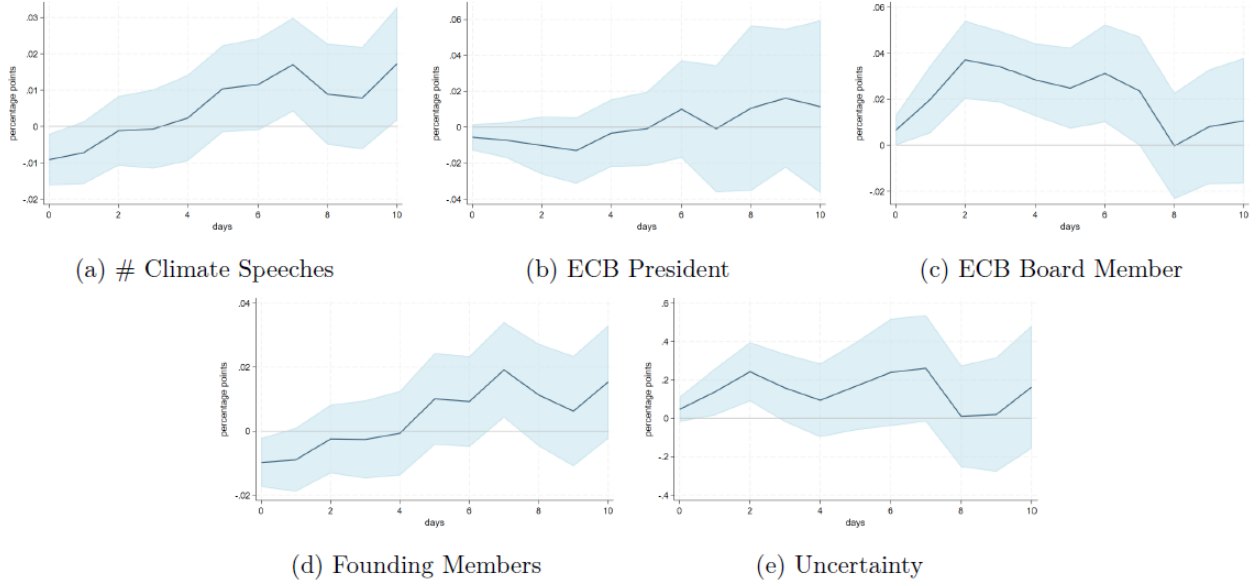


Figure C.2: Cumulative Impulse Responses of 2-year ILS to Selected Shocks (68% CI)

Note: These plots show the estimates of $y_{t+h} - y_{t-1} = \alpha_h + \sum_{m=1}^2 \lambda_{m,h} \Delta y_{t-m} + \sum_{n=0}^4 \beta_{n,h} s_{t-n} + \gamma_h x_t + v_{t+h}$, over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation expectations, and s is a) # of climate speeches, b) speeches delivered by the ECB President, c) by an ECB Board Member, d) by a representative of founding member countries, and e) the financial stability uncertainty index. The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

APPENDIX D: Estimated impacts of climate-related speeches on inflation-linked swaps, accounting for the 7-day quiet period before the ECB monetary meeting.

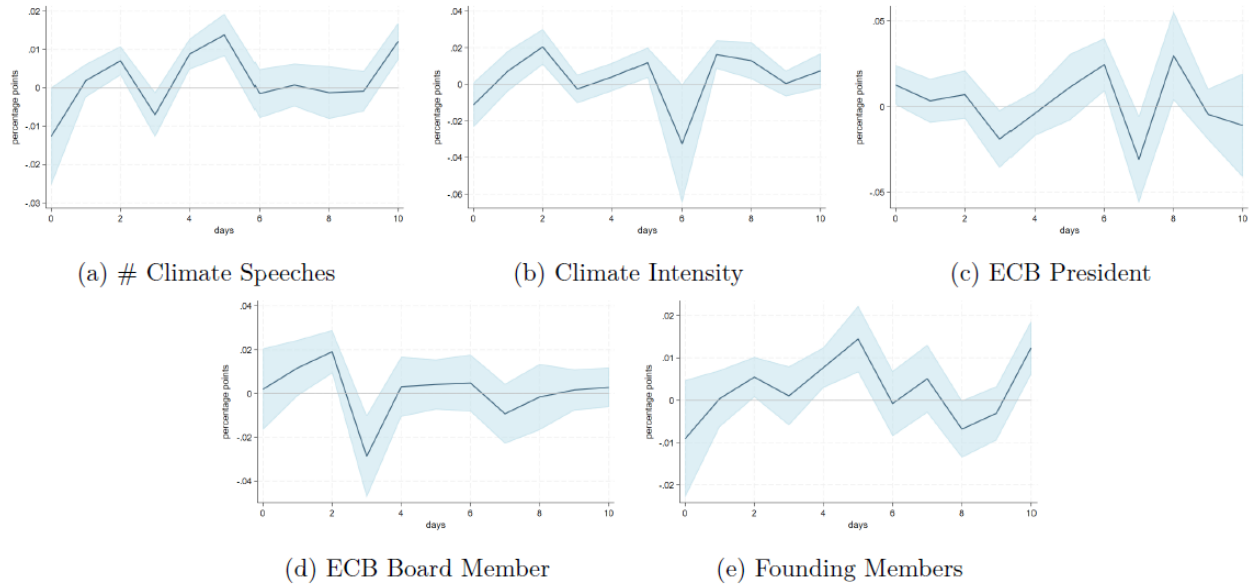


Figure D.1: Impulse Responses of 1-year ILS to Selected Shocks (controlling for the quiet period; 68% CI).

Note: These plots show the estimates of (1) controlling for the quiet period over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% and 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

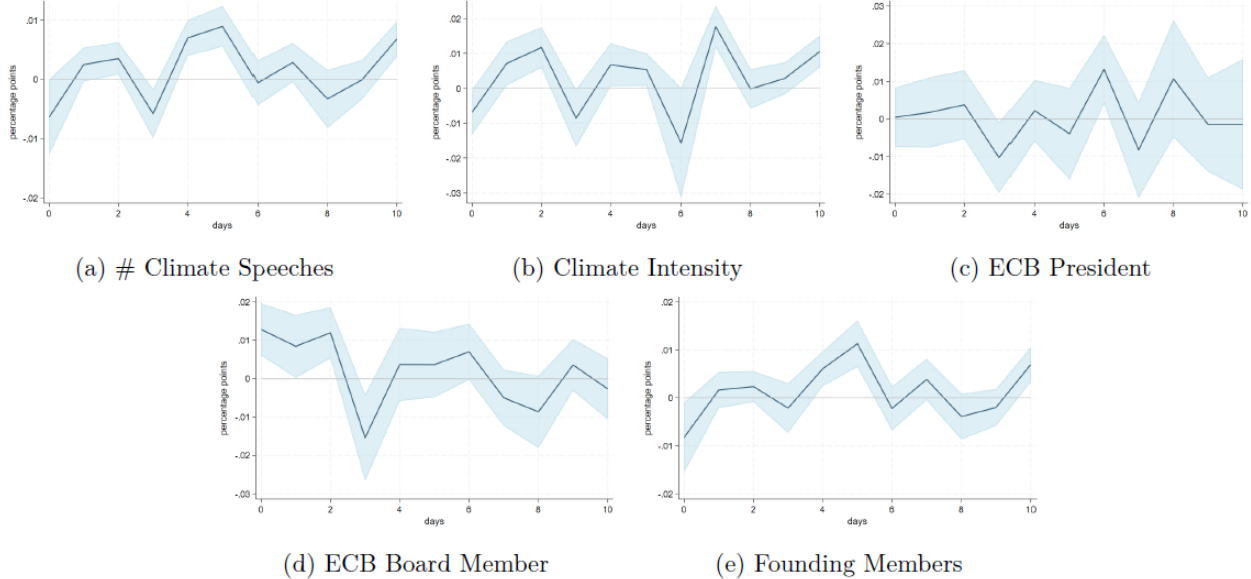


Figure D.2: Impulse Responses of 2-year ILS to Selected Shocks (controlling for the quiet period; 68% CI).

Note: These plots show the estimates of (1) controlling for the quiet period over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% and 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

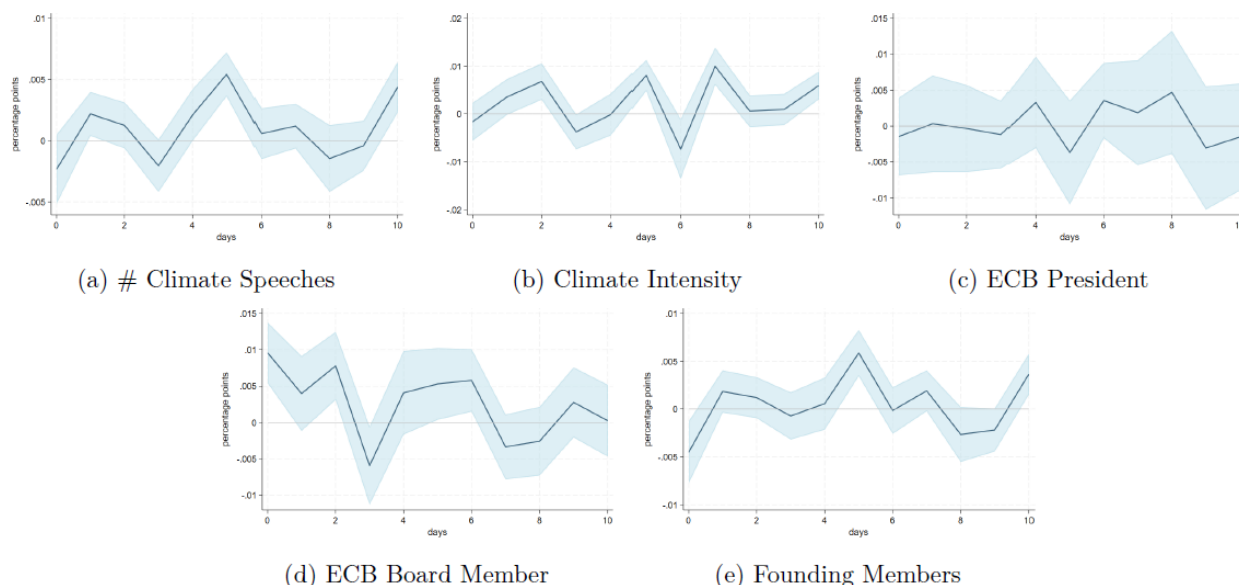


Figure D.3: Impulse Responses of 5-year ILS to Selected Shocks (controlling for the quiet period; 68% CI)
 Note: These plots show the estimates of (1) controlling for the quiet period over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% and 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

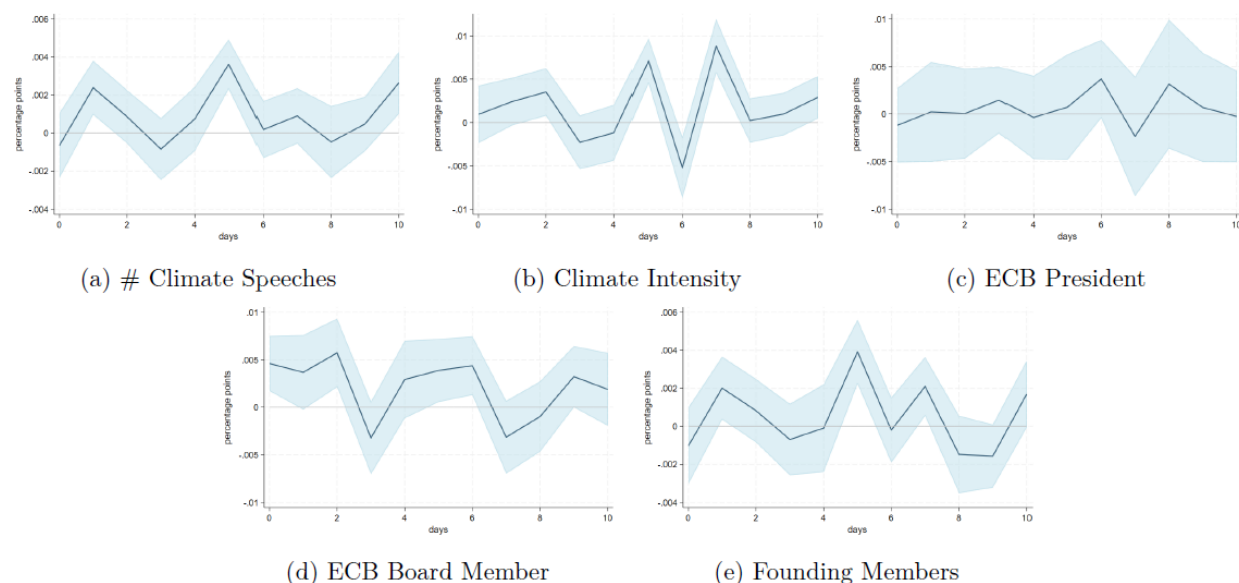


Figure D.4: Impulse Responses of 10-year ILS to Selected Shocks (controlling for the quiet period; 68% CI)
 Note: These plots show the estimates of (1) controlling for the quiet period over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation-linked swaps, and s is a) # of climate speeches, b) climate intensity, c) speeches delivered by ECB President, d) by an ECB Board Member, and e) by a representative of founding member countries. The shaded regions indicate 68% and 90% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

APPENDIX E: Vector Autoregressive Models: Estimated impacts of climate-related speeches on inflation-linked swaps

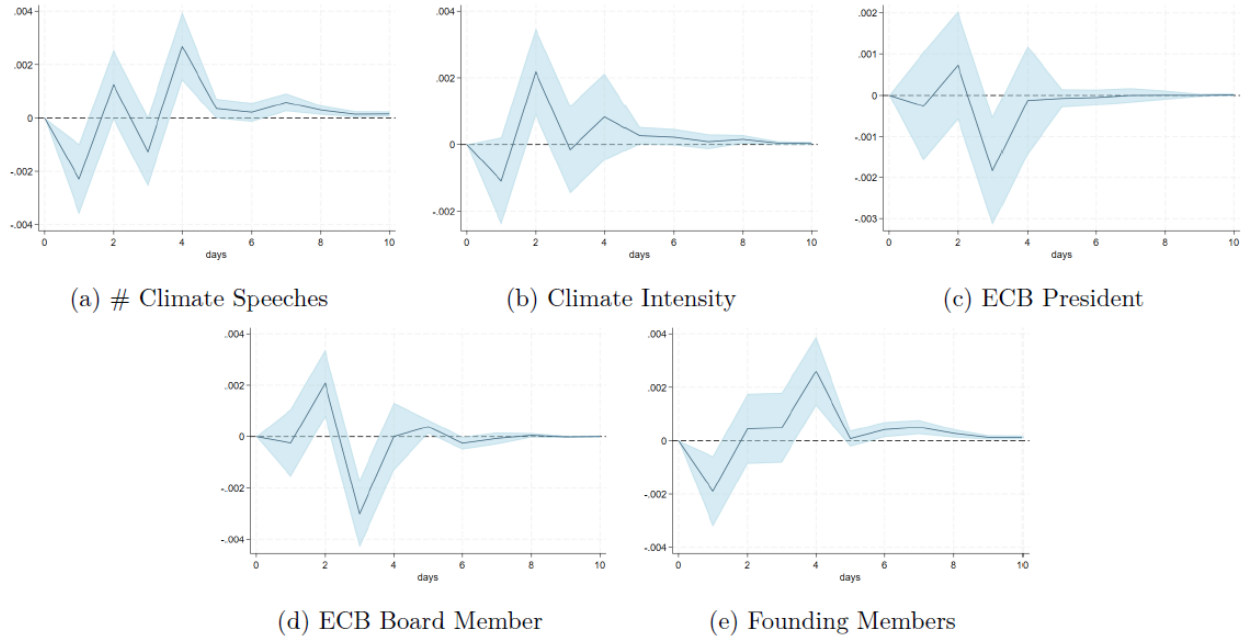


Figure E.1: VAR Impulse Responses of 1-year ILS to Selected Shocks

Note: These plots show the impulse response functions of the VAR equation $\Delta y_t = \sum_{i=1}^4 a_{11}(i) \Delta y_{t-i} + \sum_{i=1}^4 a_{12}(i) s_{t-i} + \sum_{i=1}^4 a_{1j}(i) \sum_{j=3}^6 x_{t-i} + e_{1t}$, over $h = 0, \dots, 10$ (11 days) where y is 1-year inflation expectations, s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members, and x is a vector of variables controlling for ECB meeting days and macroeconomic surprises. The shaded regions indicate 68% confidence intervals.

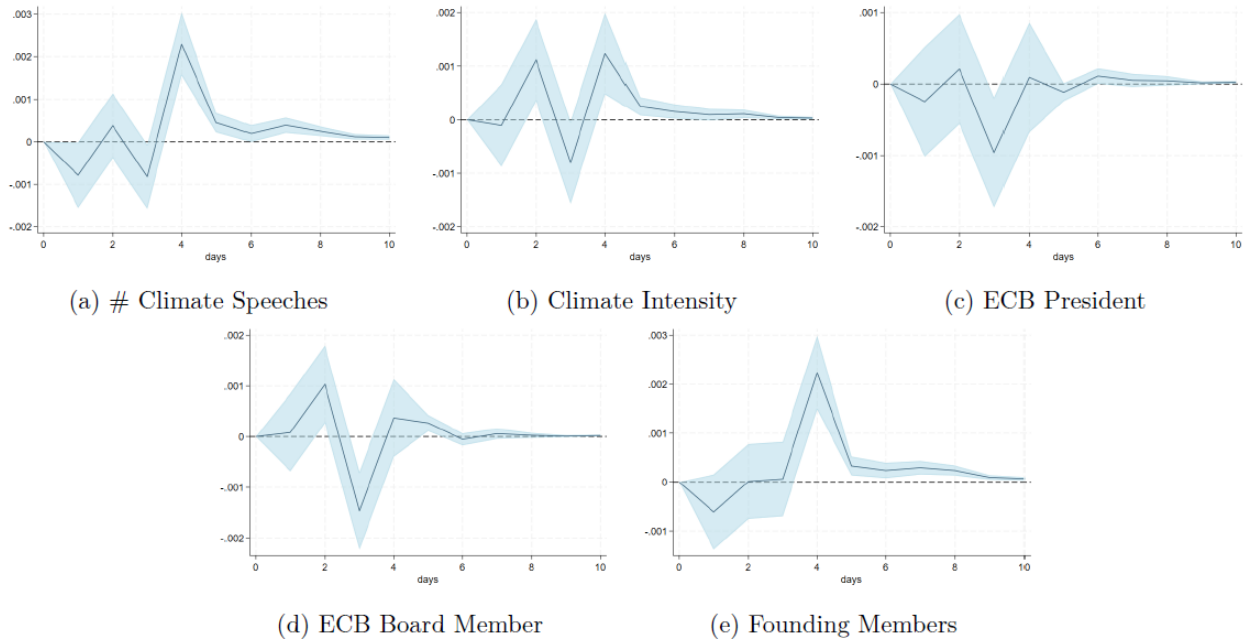


Figure E.2: VAR Impulse Responses of 2-year ILS to Selected Shocks

Note: These plots show the impulse response functions of the VAR equation $\Delta y_t = \sum_{i=1}^4 a_{11}(i) \Delta y_{t-i} + \sum_{i=1}^4 a_{12}(i) s_{t-i} + \sum_{i=1}^4 a_{1j}(i) \sum_{j=3}^6 x_{t-i} + e_{1t}$, over $h = 0, \dots, 10$ (11 days) where y is 2-year inflation expectations, s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members, and x is a vector of variables controlling for ECB meeting days and macroeconomic surprises. The shaded regions indicate 68% confidence intervals.

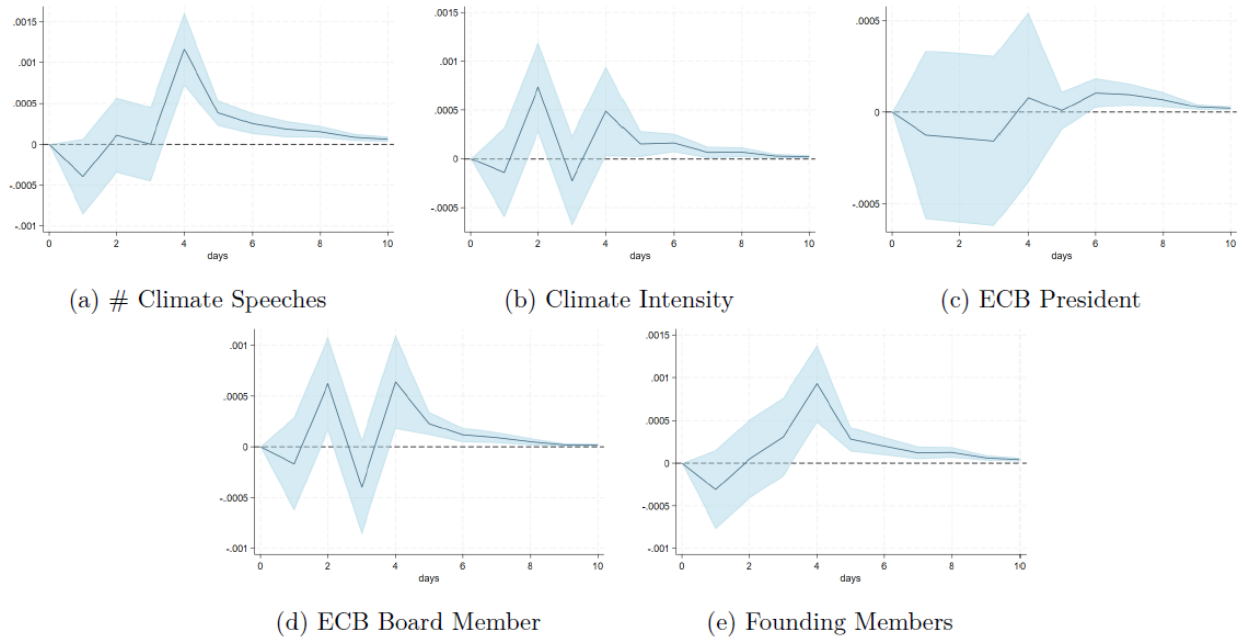


Figure E.3: VAR Impulse Responses of 5-year ILS to Selected Shocks

Note: These plots show the impulse response functions of the VAR equation $\Delta y_t = \sum_{i=1}^4 a_{11}(i) \Delta y_{t-i} + \sum_{i=1}^4 a_{12}(i) s_{t-i} + \sum_{i=1}^4 a_{1j}(i) \sum_{j=3}^6 x_{t-i} + e_{1t}$, over $h = 0, \dots, 10$ (11 days) where y is 5-year inflation expectations, s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members, and x is a vector of variables controlling for ECB meeting days and macroeconomic surprises. The shaded regions indicate 68% confidence intervals.

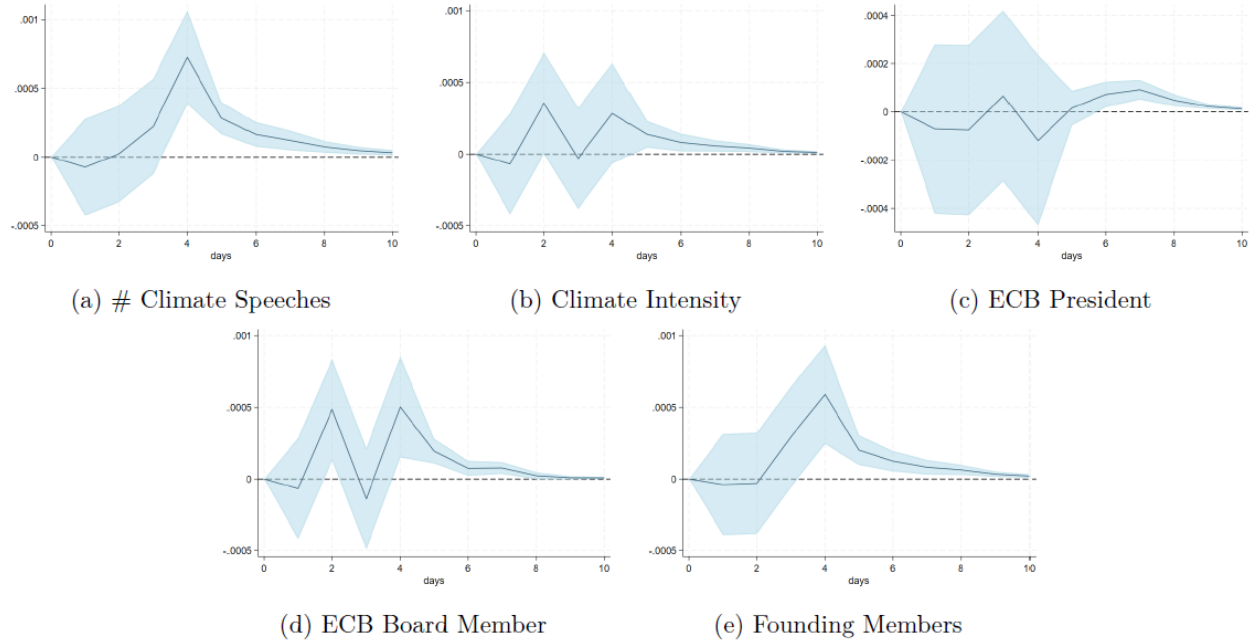


Figure E.4: VAR Impulse Responses of 10-year ILS to Selected Shocks

Note: These plots show the impulse response functions of the VAR equation $\Delta y_t = \sum_{i=1}^4 a_{11}(i) \Delta y_{t-i} + \sum_{i=1}^4 a_{12}(i) s_{t-i} + \sum_{i=1}^4 a_{1j}(i) \sum_{j=3}^6 x_{t-i} + e_{1t}$, over $h = 0, \dots, 10$ (11 days) where y is 10-year inflation expectations, s is a) # of climate speeches, b) climate intensity, c) speeches delivered by the ECB President, d) by an ECB Board Member, and e) by a representative of Founding Members, and x is a vector of variables controlling for ECB meeting days and macroeconomic surprises. The shaded regions indicate 68% confidence intervals.

APPENDIX F: Estimated impacts of climate-related speeches on inflation-linked bonds

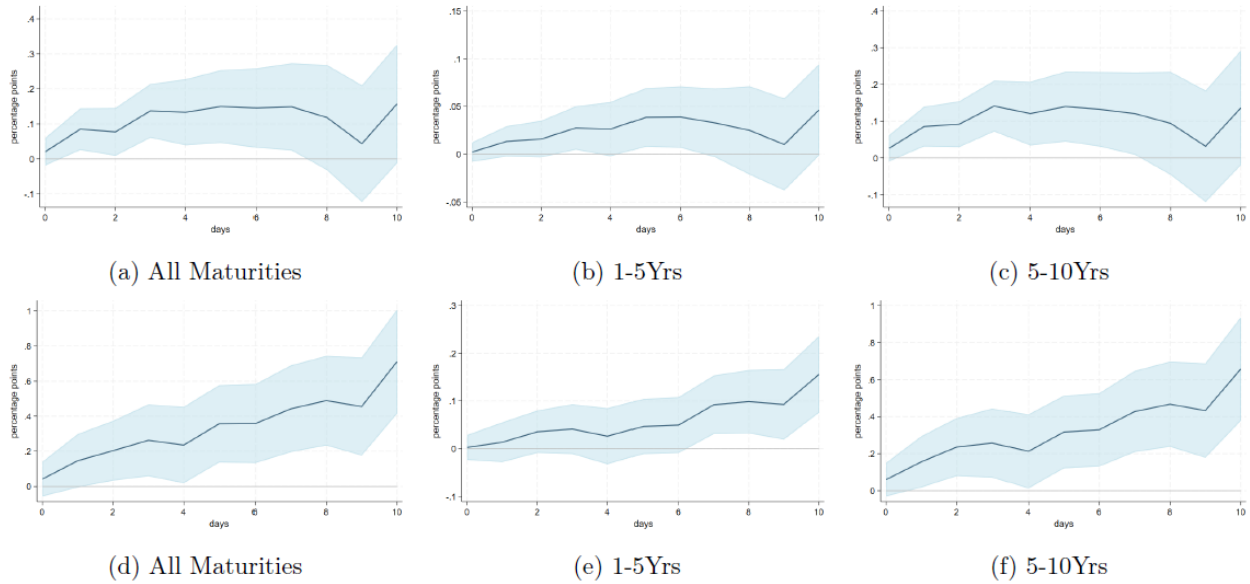


Figure F.1: Impulse Responses of Inflation-linked Bonds to Selected Shocks (68% CI, normalized speeches)

Note: These plots show the estimates of (1) over $h = 0, \dots, 10$ (11 days) where y is (a, d) Euro Govt Inflation-Linked All Maturities (BEIG1T), (b, e) Euro Govt Inflation-Linked 1-5Yrs (BEIGAT), and (c, f) Euro Govt Inflation-Linked 5-10Yrs (BEIGBT), and s is # of climate speeches in (a-c), climate intensity in (d-f). The shaded regions indicate 68% confidence intervals derived from the local projection, computed using Newey-West standard errors to control for heteroscedasticity and serial correlation.

The three indices share similar construction, and include bonds from France, Italy, Spain, and Germany. The BEIG1T index includes 37 inflation-linked bonds from these 4 Eurozone countries across all maturities. The BEIGAT index includes 11 inflation-linked bonds from these 4 Eurozone countries with maturities in the 1-5 year range. The BEIGBT index includes 10 inflation-linked bonds from these 4 Eurozone countries with maturities in the 5-10 year range.